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United Nations Development Programme

PROJECT DOCUMENT

[Lebanon]

Project Title: Improving energy security and promoting the transition to sustainable energy in Lebanon
Project Number: 01001708
Implementing Partner: UNDP
Start Date: 01 January 2024 **End Date:** 30 December 2028 **PAC Meeting date:** N/A

Brief Description
This project will provide solar energy solutions to the public sector, thus ensuring the continued provision of essential services and to bridge the energy gap that the national utility is unable to meet now. By doing so, the cost of electricity on the national budget is also lowered which would alleviate the financial burden on both the central government and on the sectors themselves, allowing them to use the financing for other essential services that are needed. On a global level, the Action is also aligned with the Lebanese unconditional National Determined Contribution (NDC) under the Paris Agreement to reduce by 20 % the Greenhouse gas emissions (GHG) by 2030, increase the renewable energy to reach 18% of the electricity produced by 2030 and a 3 % energy efficiency target by 2030. The project strategy consists of approaching the transition to renewable energy in Lebanon mainly to the public sector Parallel to these two approaches, improving capacities, policies and legislation to push the sector forward will also be tackled based on the lessons learnt and needs identified during implementation at field level of both components. Ultimately, the overall impact of the project is the improved energy security and transition to sustainable energy and resource efficiency in Lebanon. This will be reached based on the below theory of change and in the later sections, more details are provided about the detailed outcomes, outputs and activities.

Contributing Outcome (UNSDCF,CPD): UNSDCF: 1.2.b (SDG output 13.2) Reduction of CO2eq emissions, through UN support (tonnes) CPD Outcome 3: Strengthened stabilization and green recovery to reduce vulnerabilities and environmental risks, including through enhanced competitiveness and business environment of sustainability-oriented MSMEs and high potential green productive sector values chains. CPD output 3.2 (IRRF output 5.2): Transition to renewable energy accelerated capitalizing on technological gains, clean energy innovations and new financing mechanisms to support green recovery. CPD output indicator: 3.2.1. Amount of renewable energy installed annually (output indicator for UNSDCF output 1.2, IRRF 5.2.1. Increase (in megawatt) in installed renewable energy capacity with solar, wind, biomass and other technologies) Indicative Output(s) with gender marker: Output 2.2.2 Number of women engineers empowered with EU support (GEN2)

Total resources required:	USD 20,293,291.47	
Total resources allocated:		
	Donor (KfW): (Award number 1219288)	EUR 18,811,881.19 (excluding 1% UN Levy); US equivalent at exchange rate of 15 Dec. 23 (0.927)
	Government:	-
	In-Kind:	-
Unfunded:	-	

Agreed by (signatures):

UNDP	
Mr. Mohammed Salih **Resident Representative a.i.**	DocuSigned by: 4C17B57AEAA5467...
Date: 22-Mar-2024	

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I. DEVELOPMENT CHALLENGE

Background

Lebanon is enduring a severe and prolonged economic depression. According to the Fall 2021 World Bank (WB) Lebanon Economic Monitor (LEM)[1] the economic and financial crisis is potentially ranking as top 3 most severe economic collapse globally since the mid-nineteenth century. In the face of such colossal challenges, continuous policy inaction and the absence of a fully functioning executive authority to implement reforms threaten already dire socio-economic conditions and a fragile social peace.

For over two years, Lebanon has been facing compounded challenges: its largest peace-time economic and financial crisis, COVID-19 and the Port of Beirut explosion. However, Lebanon's leadership policy responses to these challenges have been highly inadequate. This is not so much due to knowledge gaps or lack of quality advice, but rather the result of a dysfunctional governance system based on vested interests, which hampers the achievement of political consensus over effective policy initiatives. With a history of a prolonged civil war and multiple conflicts— Lebanon is identified by the World Bank as a Fragility, Conflict and Violence (FCV) State - there is growing wariness of potential triggers to social unrest.

The poor quality of public services and inefficiencies in the public administration result in loss of time and resources by citizens and businesses alike and provide ample opportunities for corruption. The country's political system is designed to reinforce multi-confessional co-existence and gives way to clientelism and deeply rooted vested interests. Corruption is widespread in public and private sectors and infiltrates all levels of society, as reflected by the country's global and regional average performances scores in most areas of governance. In addition, the limited role played by civil society at policy level renders the adoption and implementation of key government reform policies difficult. The May 2022 parliamentary elections also resulted in a more fragmented Parliament, with the absence of a clear majority and growing support for alternative candidates representing civil society at the expense of traditional parties.

Real GDP has decreased by 10.5% in 2021, following a 21.4 % contraction in 2020. This represents the highest contraction in a list of 193 countries. The national currency, the Lebanese Pound (LBP), lost around 90% of its value against the US dollar since the end of 2019, which has caused triple-digit inflation as most production inputs and consumption goods are imported from abroad.

As a result, over half of the population is likely below the national poverty line, with the bulk of the labour force - paid in Lebanese Pound (LBP) - suffering from plummeting purchasing power. Increased numbers of highly qualified Lebanese, especially youth, are migrating. With the unemployment rate severely on the rise, an increasing share of households is facing difficulty in accessing basic services, including electricity, health care, water supply/sanitation and education.

Against this background, Lebanon's social and economic woes are being further exacerbated by the negative impact of Russia's war in-Ukraine, which mostly affects: (i) food (in)security, due to possible shortages in imports of crucial foodstuff and agricultural inputs; and (ii) shortages in fuel items due to the global rise in energy prices, and commodity prices more generally, aggravating Lebanon's energy crisis and inflation-depreciation spiral. In addition, negative impact is expected on the action preparedness of the security services (LAF and ISF), increasingly creating security concerns.

The electricity crisis is at the heart of a dysfunctional post-war power-sharing system and has contributed to around 46 % of the Lebanese public debt[2]. The post-war heavily subsidized electricity system has never provided full-time electricity. The Lebanese power stations owned by Electricité du Liban (EDL) run on imported polluting fuel and rely on an outdated transmission and distribution system where only 57 % of electricity produced is actually delivered, billed and collected, as the remainder is lost for technical and non-technical reasons. Unfortunately, no improvements have occurred recently, and the non-technical losses continue to increase.

[1] https://openknowledge.worldbank.org/handle/10986/36862

[2] https://www.aub.edu.lb/ifi/Documents/publications/research_reports/2020-2021/20211020_unbundling_lebanon_electricity_sector_research_paper_pdf.pdf

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EDL's unsustainable model suffers from high production cost, artificially cheap tariffs, low tariff collection rates, poor governance and lack of transparency. Since the summer of 2021, EDL has only produced up to 2-3 hours of electricity per day, forcing households and businesses to increase their reliance on privately owned, expensive and polluting diesel generators. The electricity deals signed with Egypt (for the purchase of natural gas) and Jordan and Syria (for the purchase of Jordanian-produced electricity) are yet to materialize given that the necessary funding backing the deals, to be provided by the World Bank, is conditional on reforms that are not yet being implemented.

The inability of EDL to supply sufficient hours of electricity per day and the lack of any solution in the horizon has led most households and businesses that can afford to invest in Solar PV systems to do so hastily, matter which created a boom in the local market that contributed significantly to the penetration of the renewable energies. The fast development in the demand for Solar PV also caused some inefficiencies in the market as many new contractors started working in solar technology without adequate capacity building which requires more comprehensive approaches in terms of capacity building of the labour force and implementation of the standards of the best practices of the trade.

Globally and as a result of trade and production disruptions caused by the Russian invasion of Ukraine, following the increase of the average price of Brent crude oil to USD 100 a barrel in 2022, its highest level since 2013 and an increase of more than 40% compared with 2021, prices ranged USD 81 a barrel in 2023. In Lebanon, this has pushed energy prices even higher. Diesel (mostly used for generators and transport) which was selling at 331,000 Lebanese lira per gallon - half the Lebanese minimum wage - has already undergone a 450 percent increase since the start of the Russian invasion of Ukraine.

In fact, investment in RE creates numerous job opportunities[3]. In Lebanon, driven by the significant growth of private investment, solar PV value chain has added at least 600 new jobs and the number of the companies has grown from 13 companies in 2010 to more than 50 companies by the end of 2022[4]. The UNDP study have shown that investment in solar PV system creates various local "direct" and "indirect" jobs in Lebanon[5]. Direct jobs include construction works, the design of the solar PV system, and other business activities such as financing, admission, legal services, consultation and planning. In addition, along the solar PV value chain, the demand for wiring, cabling, legal services etc. creates indirect jobs in the respective economic sectors. It is estimated that each $1 million investment for 1 MW of solar PV installation creates at least 11 direct jobs (Full-Time Employment equivalent, FTE) and 20-25 indirect ones, amounting to 30-45 FTE jobs along the said value chain. This indicates that further acceleration of investment in the installation of solar PV will expand job-demand in the sector. Moreover, it is expected that once the markets for commercial solar thermal, biogas and EE measures (notably heat recovery) are initiated and promoted, new employment opportunities will be created in a similar manner.

Since the current Lebanese energy service providers (EDL and private generator) are unreliable, expensive and dependent on imported fuel, there are untapped market opportunities for RE and EE solutions albeit innovation, entrepreneurship, and technology transfer are necessary to open up these markets. Even though innovation and entrepreneurship have great potential to promote new energy markets, innovation via research and development (R&D) is generally very weak due to insufficient and irregular funding. Strengthening linkages between university research and market/industrial needs can address this innovation challenge. In terms of entrepreneurship, the World Bank identified the most significant success factor for long-term success (hiring employees over time and continuing to do) of Lebanon's startup is additional work experiences[6]. Participation in acceleration programs has no significant effect on long-term hiring probability and serial entrepreneurs in Beirut are less likely to hire consistently. These issues imply that it is very critical to get buy-in from private companies for innovation and entrepreneur program and nurture sector-specific market/industrial insights within human capital for the sector (i.e. entrepreneurs, researchers, students) in order to promote innovation and entrepreneurship. Technology transfer can also

[3] IRENA (2018). Renewable Energy and Jobs – Annual Review 2018.

[4] LCEC List of Qualified Solar PV Companies (January 2023).

[5] UNDP (2019). Prioritization and Assessment of Value Chains within the Renewable Energy Sector in Lebanon.

[6] WB (2017). Tech Start-up Ecosystem in Beirut: Findings and Recommendations.

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make up for lack of R&D funding and available commercialized RE/EE technologies by introducing internationally viable solutions as proven by past UNDP's demonstration projects[7].

Regarding the Hydro sector, Lebanon is blessed with relatively rich water resources in terms of precipitation, equalling on average approximately 823 millimetre (mm) annually, however with significant annual variation between 600 - 1000 mm in the coastal area, 900 - 1,700 mm in the Lebanon mountain range, 500 - 900 mm in the eastern mountain range, and from 200 - 900 mm in the Bekaa Valley (JICA, 2003). Much of this precipitation is, however, lost to evaporation, to neighbourhood countries, and to the Mediterranean Sea, while only a relatively small percentage is gained to controllable underground and surface water. Furthermore, statistical analysis of annual precipitation based on 80 years observation shows a decreasing tendency starting from the year 1970 (JICA, 2003).

All the hydropower plants were constructed pre-1970. Lebanon has an approximate installed hydropower nominal capacity of 280 MW, with the Litani River Authority (LRA) managing approximately 190 MW of this capacity along the Litani-Awali River. Currently, the energy produced from the hydro plants constitutes 4.5% from the total production (MoEW, 2010). The Ministry of Energy and Water (MoEW) 2010 Policy Paper for the Electricity Sector indicated that hydraulic power production's share in the electricity mix would be increased by at least 120 MW through "maintenance, rehabilitation and/or replacement of existing hydro plants, and additional capacity". In addition, the Policy Paper also called for the encouragement of "all individual and private initiatives to produce hydropower, even micro-hydro".

In this context, MoEW prepared a Master Plan for the Hydroelectric sector development in Lebanon in collaboration with the French Agency for the Development (AFD) and the consultants Sogreah/Artelia. This Master Plan investigated the hydroelectric development potential along all the Lebanese rivers that are entirely within the Lebanese territory and identified 32 new sites with the following potential:

- around 263 MW (1,271 GWh/y) with a cost of $ 667 million (in run of river scheme)
- around 368 MW (1,363 GWh/y) with a cost of $ 772 million (in peak scheme)

Of those sites, 25 are economically viable with minimum selling tariff inferior to 12 US$ cents per kWh, corresponding to the following potential:

- around 233 MW (1,126 GWh/y) with a cost of $ 560 million (in run of river scheme)
- around 315 MW (1,217 GWh/y) with a cost of $ 665 million (in peak scheme)

The MoEW has also investigated, with UNDP-CEDRO and various public establishments, the possibilities for the rehabilitation of the existing 280 MW hydro fleet which were installed between the 1920's and the 1960's and have all exceeded their technical lifetime. Finally, the MoEW, in collaboration with UNDP-CEDRO, has explored the possibilities for the Micro-Hydro development from non-river sources on 13 pilot sites.

In addition to the hydro rehabilitation and upgrade component, this project will be investigating the integration of hydro pumped storage to the electricity system of Lebanon for peak shaving and increased RE deployment in the energy mix as well as the use of hydro units as capacitors or the grid stability improvements.

Relevance of the Action

This project will provide sustainable energy solutions to the public sector, thus ensuring the continued provision of essential services and to bridge the energy gap that the national utility is unable to meet now. By doing so, the cost of electricity on the national budget is also lowered (Estimated Economic savings Assuming LCOE is 0.27 USD/KWh will be around 1.8 MUSD per year) which would alleviate the financial burden on both the central government and on the sectors themselves, allowing them to use the financing for other essential services that are needed. The project will focus on providing RE/EE interventions to the remaining public hospitals and the only public university in Lebanon together with the rehabilitation, upgrading and retrofitting of the existing hydropower plants at Nahr El Bared.

[7] The opportunities in price reduction and increase in PV service providers witnessed from the EU funded CEDRO 4 project.

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

The project supports the public sector by implementing RE/EE projects. Utilizing training initiatives that aim at enhancing the quality of renewable energy system offerings in the market, along with policy measures to reduce risks in the sector, represents a strategy for expanding the market and guaranteeing the sustainability of high-quality systems. On a global level, the project is also aligned with the Lebanese unconditional National Determined Contribution (NDC) under the Paris Agreement[8] to reduce by 20 % the Greenhouse gas emissions (GHG) by 2030, increase the renewable energy to reach 18% of the electricity produced by 2030 and a 3 % energy efficiency target by 2030. It also aligns with the national agenda: the Ministry of Energy and Water's Electricity Policy Paper and Lebanon's Recovery, Reconstruction and Reform Framework[9] (3RF) on energy which is endorsed by both the Government, the World Bank, the United Nations and the European Union. The 3RF is a people-centred recovery and reconstruction framework to put Lebanon on the path of sustainable development focusing on essential actions, such as addressing the urgent needs of the most vulnerable populations and small businesses affected by the crisis.

In line with the European Commission policy priorities[10], this action will support the external dimension of the European Green Deal and in particular the uptake of sustainable energy solutions, circular economy and green recovery in both the public and private sector. In the private sector, this action will support the industries to use new innovations and digital solutions and adopt circular economy practices and value chains. In the public sector, it will support pressing needs to increase access to affordable electricity and comprehensive energy management solutions (demand-side management, energy efficiency and small-scale renewable systems), and eventually improve the provision of basic public services notably in the education and defence sector.

This project will notably build on the success of previous UNDP energy projects to further support. UNDP has previously implemented a Solar for Health project with KfW funding and therefore has experience in working with hospitals. Earlier phases of renewable energy projects started in 2006 where solar systems were designed and installed in various types of public institutions including public schools, municipal building, ministerial building, NGOs and others. Experience and lessons learnt span from technical design knowledge that has been acquired to knowledge of equipment and specifications that are best suited for the Lebanon and public sector requirements, to insight into the public sector's electricity structure and needs as well as the capacity of local market suppliers, to name a few. The approach is based on existing and successful programmes in Lebanon to accelerate the green energy transition while creating sustainable jobs.

The successful Solar for Health project implemented under the "LHSP – Response to the impact of COVID-19 in Lebanon" will also serve as a steppingstone for the current project in terms of upscaling the existing interventions at the 15 public hospitals and expanding the intervention to cover all remaining public hospitals . The project will also provide RE/EE measures to the only public university in Lebanon that will in turn ensure the access to in-person learning in facilities with unhindered and proper power supply.

In addition to the above, the project also plans on rehabilitating, upgrading and retrofitting the existing hydropower plants at Nahr El Bared to increase the overall nominal power by 10% which will in turn increase the total energy fed to the grid and benefit up to 3,600 persons in the area.

[8] https://unfccc.int/sites/default/files/resource/Lebanon%20BUR4%202021.pdf

[9] The 3RF provides a roadmap to operationalise the finding of the Rapid Damage Needs Assessment (RDNA) and other assessment carried out in response to the devastating port explosion where a large amount of ammonium nitrate caused at least 210 deaths, reported missing persons, and more than 6,500 injured.

[10] https://www.europarl.europa.eu/RegData/etudes/BRIE/2020/646148/EPRS_BRI(2020)646148_EN.pdf

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

II. STRATEGY

UNDP is the lead agency on policy support, advocacy and the implementation of sustainable energy projects in Lebanon. UNDP has been working in this sector, together with the Government of Lebanon, since 2004. In addition, UNDP brings the following main comparative advantages:

- Global and regional expertise and experience in promoting sustainable energy strategies and projects
- Proven track record of working with the Ministry of Energy and Water and others, local communities, the private sector and public institutions on the implementation of sustainable energy solutions using various types of technologies
- Extensive experience in integrating sustainable energy solutions into emergency response, recovery and development agendas in Lebanon and the region
- Application of gender and environment considerations in all project implementation
- Solid experience in implementing infrastructure projects

UNDP has already implemented over $25 million in funding related to solar energy systems with funding from several donors since 2008, including but not limited to the Spanish Government, European Union, Governments of Netherlands, Japan and Germany, and the Global Environment Facility (GEF) to name a few. UNDP's programme on renewable energy and energy efficiency started in 2004 and continues to today. The activities implemented range from creating new value chains in renewable energy and energy efficiency through design, procurement and implementation of such technologies with private sector suppliers (both local and international) as well as the preparation of technical publications on renewables, the analysis of the local market, policy support, assistance to the Ministry of Energy and Water on the preparation of related technical legislation, communication and awareness raising and others.

This project aligns with the new UNDP Country Programme Document (CPD) for 2023 – 2025, specifically Priority 3 which aims at "Building on the crisis response, support the country's longer-term aspirations towards green and inclusive development". This in turn specifically supports the Sustainable Development Goals 5, 7, 8, 12, 13, 15 and 17. Within this priority, UNDP commits to develop capacity in institutions to implement and oversee support gender-responsive interventions in the sustainable energy value chain, solid waste management, water and wastewater management and green production models aiming to:

- Reduce environmental degradation and protect natural resources; seek longer-term solutions to reduce pollution and to natural resource efficiency,
- mitigate and adapt to the climate crisis, and
- promote sustainable land management and ecosystem approaches.

The project strategy consists of approaching the transition to renewable energy in Lebanon at three levels: one is to provide and improve access to the public sector, the second is to support the private sector to upscale solar energy supply to greening recovery and creating jobs and the third is to build capacity and provide technical policy support to the government in this area.

Ultimately, the overall impact of the project is to contribute to the transition to sustainable energy in Lebanon. This will be reached based on the below theory of change and in the later sections, more details are provided about the detailed outcomes, outputs and activities.

Theory of Change (TOC)

If (Activities)

- Deployment of sustainable energy solutions across the public service sectors
- Upgrading the hydropower capacity in Lebanon and preparing a hydropower strategy
- Augmenting the capacity, raising awareness, and providing comprehensive technical assistance public entities

Then (Outputs)

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

- Enhanced energy savings and access to electricity through renewable energy and energy efficiency installations
- Elevated public service delivery due to reliable energy supply and cost savings due to cheaper energy supply
- Boosted job opportunities within the sustainable energy sector
- Upscaled capacity and expertise public entities to embrace sustainable energy solutions
- Increase in hydro-energy supply at national level

Then (Outcomes)

- Strengthened energy security and resilience in Lebanon
- Enhanced public service delivery driven by reliable and sustainable energy supply, in the health and tertiary education sector
- Increase in job opportunities and economic growth via the energy sector
- Improved market landscape for renewable energy and sustainable practices in Lebanon

Then (Impact)

- Sustainable and resilient economic development in Lebanon
- Improved quality and reliability in public service delivery
- Decreased energy dependency and bolstered energy security
- Improved gender participation in economic activities

Assumptions (Risk Management)

- Active participation and adoption of sustainable energy practices and standards by industries, SMEs, public entities, and start-ups
- Availability and allocation of ample resources and funding for the project execution
- Stable political and economic landscape facilitating the seamless implementation and progression of the project
- Consistent collaboration and endorsement from local partners and stakeholders

The project's Theory of Change (ToC) aligns with the UN Strategic Development Framework and the UNDP Country Programme Document by providing a definitive pathway to sustainable development, enhancing public services, generating employment, and bolstering energy security, all of which are core constituents of both strategic plans.

The chosen strategy of fostering close alliances with the private sector and public entities is grounded in UNDP's successful track record with analogous initiatives in Lebanon and globally. Existing evidence validates that such a participatory approach amplifies the adoption of sustainable energy solutions, culminating in the desired transformative changes.

Gender

The challenge of achieving sustainable development is intricately linked with the need for gender equality. Unfortunately, a gender gap exists in renewable energy sector in Lebanon, particularly in terms of work opportunities for women in the energy value chain.

Building on UNDP's previous work to bridge the gap in the field of gender and energy as well as climate change action, improving gender participation will be mainstreamed across the various activities as possible in this Action. This will include the promotion of women participation in the solar energy training programmes and the encouragement of their active participation to encourage women engineers, technicians and installers in the sector. Gender considerations will also be integrated in the innovation activities to encourage young women entrepreneurs to participate in sustainable energy and climate start-ups and ideas.

Furthermore, the project will engage with policymakers, advocating for gender-inclusive energy policies and frameworks. By highlighting the social, economic, and environmental benefits of gender-responsive energy solutions, it aims to create an enabling environment for change.

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

Ultimately, the project strives to create a future where the involvement of youth and women in the sustainable energy and climate challenge is fully integrated. By championing gender equality in the renewable energy sector, it aims to unlock the potential of women, promote economic empowerment, and pave the way for a more sustainable and equitable world.

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

III. RESULTS AND PARTNERSHIPS

Expected Results

The Overall Objective (Impact) of this action is to improve the energy security supply to the selected beneficiaries and to contribute the transition to sustainable energy in Lebanon.
The Specific Objectives (Outcomes) of this action are to:

Outcome 1: Increased quality and access to public services through sustainable energy solutions

Output 1.1 Increased energy savings and access to electricity in public service facilities

This first outcome will support the uptake of renewable energy and energy efficiency equipment in public sector buildings, hence contributing to both energy transition and a more efficient provision of public services and to bridge the energy gap using clean technologies.
Also, it will support the increase of renewable energy generation by the electricity public utility (EDL), hence contributing to both energy transition and a more efficient provision of public services and to bridge the energy gap using clean technologies.

- **Activity 1.1 Sustainable Energy for Public Services**

  Public service entities, whether at national or local levels, are struggling to operate due to the low level of electricity supply from the public electricity company Electricité du Liban (EDL) (2-3 hours per day on average). This is affecting the proper operation of schools, hospitals, Ministries, municipalities and all other public service providers all over the country, and therefore impacting the quality and continuity of public service delivery. The increased reliance on fuel generators has caused additional costs that the institutions concerned are unable to cover or pass on to final users, hence leading to major disruptions in the proper functioning of the State and local administrations.
  The sharp deterioration in the provision of basic public services due to the electricity crisis have concrete implications for the functioning of schools, hospitals and other basic services leading to partial closure of schools or health care facilities in certain extreme cases. According to the World Bank, the collapse of public service provision is expected to have serious and long-term consequences in terms of human capital loss for the country11. Providing the public sector with a more reliable source of electricity is therefore crucial to improve the delivery of essential public services, in particular for the most vulnerable population. Within this activity, several types of sustainable energy for public services will be explored including solar for the healthcare and education sectors.

  Within the 'Solar for Health'-KFW funded project, UNDP has developed and published the "Sustainable Energy Strategy for the Health-care sector in Lebanon" to enhance the energy performance of Lebanese public hospitals meeting the 2030 renewable energy and energy efficiency targets for Lebanon. Furthermore, and as part of the strategy, the World Health Organization (WHO) assessed hospital medical equipment to identify their energy characteristics and needs, using a combination of desk review and field visits for a mix of private and public hospitals. Based on this, WHO developed energy saving criteria for medical equipment per hospital department, the outcomes and results of which are included in this assessment.

  UNDP has already completed, with earlier financing from the Government of Germany, through KfW, the installation of renewable energy and some energy efficiency measures in 15 public hospitals, identified by the Ministry of Public Health, but to a partial extent. The current interventions will focus on the remaining public hospitals that were not part of the that project and to complete, whenever feasible, the remaining interventions in the 15 hospitals. The targeted public hospitals across the country may include the Zahle, Baabda, Tebnin, Ehden, Hasbaya, Jezzine, Kana, Kartaba, Marjaayoun, Mays El Jabal, Orange Nassau, Rachaya, Tyre, and Tannourine. Renewable energy and energy efficiency measures in the public hospitals will be covered according to the energy needs of each hospital that will be identified through energy audits, and the most energy efficient measures that

[11] World Bank, Spring 2021

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

have not been installed yet will be implemented as well in the 1st batch of the 15 public hospitals where they will be revisited. Below is a summary of the main interventions to be included in the project and that may have longer pay-back periods in the Energy Audit recommendations but may be selected in all cases by the project given that new types of energy efficiency or renewable energy interventions are to be promoted within the scope of this project:

- Solar Photovoltaic Systems (with or without batteries backup depending on the needs on site – batteries could be included for critical systems)
- Solar Water Heaters (or additional PV panels in case the hospital has electric water heaters instead of a central heating system)
- LED Lighting Retrofit
- HVAC Systems Upgrades

The measures could include the HVAC system upgrades wherever feasible and will not only be based on the analysis of the energy audits but also on-site visits to the hospitals to assess the heating and cooling needs as well as the infrastructure of each facility to determine whether it is viable to implement such measures. Installing or performing HVAC system upgrades or new installations in operational hospitals poses several significant risks and operational challenges. Hospitals require a stable and controlled environment to ensure patient care and safety, making any significant disruptions to HVAC systems highly critical. Shutting down HVAC systems for maintenance or upgrades can lead to temperature and humidity fluctuations, which can negatively impact sensitive medical equipment, patient comfort, and the effectiveness of infection control measures. Moreover, airborne pathogens may become more difficult to control in an environment with compromised ventilation. The interventions must be carefully planned and coordinated to minimize interruptions and disruptions, provide temporary solutions if needed, and ensure the highest level of infection control during and after the works to safeguard the well-being of patients and staff. Additionally, emergency response plans should be in place to address any unforeseen HVAC system failures during the installation process. The HVAC-related interventions in the originally visited 15 public hospitals could include but are not limited to the following:

- Replacement of split AC units by variable refrigerant flow systems:
  - VRF systems are highly energy-efficient compared to traditional split AC units as they use variable speed compressors that adjust the refrigerant flow according to the cooling or heating demand of each zone. This precise control allows the system to operate at partial load conditions, resulting in energy savings.
  - The systems also provide individual zonal control, meaning each indoor unit can be controlled separately, allowing different temperature settings in different areas.
- Installing Heat Pump systems where feasible and including but not limited to the below different options:
  - Air-Source Heat Pump (ASHP): extract heat from the outdoor air and transfer it indoors to provide heating or cooling. They are the most common and cost-effective type of heat pump for residential use.
  - Ground-Source Heat Pump (GSHP) or Geothermal Heat Pump: GSHPs use the stable temperature of the ground or a water source (like a well or pond) to exchange heat with the building. They are highly efficient but require a more extensive installation.
  - Water-Source Heat Pump (WSHP): WSHPs extract or reject heat to or from a water source, such as a well, lake, or cooling tower. They are commonly used in large commercial buildings.
  - Hybrid Heat Pump: Hybrid heat pumps combine a conventional heating system (like a gas or oil furnace) with an electric heat pump. They offer versatility and energy savings by switching between the two heat sources as needed.

  These types of installations are very construction-heavy especially in facilities or buildings that are already constructed (they are usually installed during the initial construction phase) given that they entail a lot of plumping and electric requirements that may be disruptive or result in extensive civil works. Therefore, the time and impact for the implementation of these systems in hospitals that are already operational will be taken into account during the selection process/at inception phase. The acceptance of the beneficiaries (the hospital management) is a pre-requisite.

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

- Investing in new heating boilers and steam boilers in hospitals will lead to improved operational efficiency, enhanced safety, and reduced environmental impact, ultimately supporting the well-being of patients and staff while minimizing costs:
    - The new boilers are highly energy-efficient, offering improved fuel combustion and reduced energy consumption. This can lead to significant cost savings for hospitals, especially considering the substantial energy requirements of large healthcare facilities.
    - Energy-efficient boilers also have a lower carbon footprint, as they produce fewer greenhouse gas emissions compared to older or less efficient systems. By upgrading to new boilers, hospitals can contribute to environmental sustainability and align with their green initiatives.
    - Upgrading to new boilers also ensures that hospitals benefit from the latest technologies and engineering advancements, resulting in more durable and long-lasting systems. This minimizes the need for frequent repairs and replacements, reducing downtime and maintenance costs.
- Proper maintenance of existing solar water heating systems ensures optimal performance, energy efficiency, cost savings, and environmental sustainability. It also promotes safety and helps protect the investments by prolonging the system's lifespan and maintaining compliance with warranty requirements:
    - System maintenance helps ensure that the solar water heating system operates at its optimal performance level, extending its lifespan. Well-maintained solar water heating systems operate more efficiently, which translates to increased energy savings. This also ensures that the solar water heating system continues to provide a consistent and reliable supply of hot water.
    - Properly maintaining solar water heating systems will lead to cost savings, reduction of greenhouse gas emissions and reliance on fossil fuels, promoting a greener and cleaner environment. The intervention will also address any potential safety hazards, such as faulty electrical connections or leaks, ensuring the well-being of personnel and preventing accidents.

The exact interventions in sustainable energy, both through energy efficiency and reduction through renewable energy, will be specified once detailed technical assessments on buildings are undertaken and based on the budget available. Any remaining budget will be used to install sustainable energy solution in primary healthcare centres which will be selected and prioritised with the Ministry of Public Health and coordinated with other international partners that are implementing similar activities. It is also worth mentioning that addressing the extended payback period associated with energy-efficient (EE) measures is a crucial aspect of our project's new approach. While EE measures often require a more extended timeframe to deliver returns on investment, we recognize the importance of their environmental and financial benefits over the long run. In our approach, we will emphasize a holistic approach that considers not only immediate costs but also the significant savings and sustainability advantages EE measures offer over the lifecycle of the installed systems. By adopting this perspective, we aim to demonstrate that the initial investment in EE technologies is an investment in the hospital's future, encompassing reduced energy bills, enhanced environmental responsibility, and improved patient care through a stable and efficient healthcare environment. Our project will prioritize a comprehensive cost-benefit analysis that accounts for the extended payback periods, ultimately making a compelling case for the inclusion of EE measures in the hospitals' systems.

In addition to the above, one of the targets of this component is the education sector. One of the main pillars of the education system in Lebanon is the Lebanese University (LU) with 19 faculties and 3 doctorate schools spread across 76 branches all over the country. The LU, through all its branches, caters to nearly 70,000 students with female students making up more than half of the student numbers.

The provision of sustainable energy to the LU is vital to carry out in-person learning. This is important not only because it provides access to education, but also given the electricity and internet availability situation of university students at home. The detailed specifications of interventions, both through energy efficiency and reduction and through renewable energy, will be designed in the selected

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Lebanese University campuses and faculties, based on energy audits that will be carried out by UNDP once the project is initiated.

Furthermore, UNDP had been previously informed that there are potentially other donors/projects such as the World Bank that might already supporting the LU so, a coordination meeting was conducted on the 13th of July 2023 with the Lebanese University and World Bank representatives, during which the WB's intervention was discussed to ensure complementarity with UNDP's projected interventions upon project initiation. The WB's 2M USD project will focus on providing solar PV systems in the Hadath, Fanar and Tripoli campuses with the budget divided as follows: 50% Hadath, 25% Fanar and 25% Tripoli. UNDP's RE/EE interventions will be conducted in a way to supplement them with top-ups for RE activities where possible and EE activities throughout the 3 campuses and after detailed energy audits are completed.

It is also worth highlighting that net meters will be provided for the facilities benefitting from RE/EE interventions to ensure excess energy generated during sunny periods is fed back into the grid, which will assist in offsetting the utility costs. This promotes energy savings, environmental sustainability, and potential revenue generation through feed-in tariffs, making it a win-win for the beneficiaries and the EDL grid.

Increased awareness and capacity building for public servant employees on sustainable energy management and climate change is crucial for effective policy implementation, decision-making, and promoting sustainable practices within public entities. Capacity building and trainings will be implemented throughout the installation of the renewable energy systems for each facility and beneficiary.

Outcome 2 Sustainable Energy for Electricity Supply
Output 2.1 Increased sustainable energy sources for electricity supply

In the 1960s, more than half of Lebanon's electricity needs where met through hydropower, placing Lebanon back then as a pioneering country in terms of renewable energy supply. Today, this hydropower resource has been bereft of attention and investment since the start of the civil war in the 1990's, leaving most of the current existing hydropower plants in need of major rehabilitation and upgrades (all the current hydro turbines date back to the 50s and 60s), and has left many possible new hydropower potential plants unconstructed. The hydropower sector can swiftly play an important role in meeting the increasing demand and reduce the risks of exposure to oil prices fluctuation, since significant local hydropower experience and infrastructures already exists.

The proposed potential activities include the rehabilitation, upgrading and retrofitting of the existing hydropower plants at Nahr El Bared. The El Bared company, formerly a private concession and now fully owned by the electric public utility and sole off-taker EDL, operates two hydropower plants, namely: The El Bared 1, which is located at the level of Nahr El Jdaïde. The El Bared 2 which is located on the Nahr Abu Moussa. The installed capacity of El Bared 1 is 13.5 MW and the capacity of El Bared 2 is 3.7 MW. Both HPPs have been constructed in 1936 and limited maintenance has been conducted during the last decades leading to lower energy outputs than initially installed. The El Bared hydropower facility of Bared 2 have been idle since June 2022 as a result of lack of maintenance whereas the Bared 1 HPP is still partially in operation with derated capacity.

Following a rehabilitation action, it is estimated that the total energy output can be increased by 15% with respect to the baseline values for energy production. The available records show that the El Bared hydropower facilities have been operating with a capacity factor of 63% during the last decades with an average yearly production of 40.994 GWh/year. The potential electricity supply that could be added to the network from the El Bared Hydro plants as a result of a rehabilitation can go up to 6,149 MWhrs, whereas the average amount of energy that will be restored as a result of putting back the units in service in safe operation at their full potential is 40,994 MWhrs. Thereby the total energy that will be delivered to the grid as a result of the rehabilitation intervention is considerable as it would cover the needs of an estimated population of 27,000 persons.

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The proposed intervention on El Bared 1 and El Bared 2 hydropower facilities falls within the first Cluster of the planned road map of hydropower development for the El Bared basin already identified by the Ministry of Energy and Water as follows (ref. WB study D4):

- Cluster 1: El Bared 1 and El Bared 2 (rehabilitation works for existing sites)
- Cluster 2: Bared, El Mara, El Ouatie, Qarn (development of new sites)
- Cluster 3: Sir, Qattine, Mechmech (development of new sites)

The rehabilitation of the two existing HPPs require detailed feasibility studies which will, inter alia, delineate the framework of the water rights of use. Typical content of this action include:

- Site survey and detailed assessment of hydrology, environmental impacts (collection of all available data; measurement of missing information; analysis of data)
- Definition of most attractive option in terms of energy yield, cost, environmental impact etc.
- Engineering design allowing cost estimates (bills of quantities) for the rehabilitation of headworks (intakes, canal, penstocks, turbines, powerhouse etc.)
- Economic and financial analysis
- Risk analysis and sensitivity analysis
- Preparation and submission of environmental impact assessment (as stipulated by the national legislative framework).
- Securing relevant permits and licenses

On another note, the El Bared company owns land in the vicinity of the existing El Bared hydro facilities and so this will be assessed for the potential development of solar PV farms that can be integrated with the hydropower plants to optimize the energy supply throughout all the seasons of the year. Such solar PV farms can be deployed if found feasible when more funds will be available.
For a given level of hydropower development, the analysis on a basin level allows for the identification of the optimal combinations of projects that enable adjustments of other policy parameters such as water management, environmental mitigation measures, social policy etc. Given that the Ministry of Energy and Water is the public sector entity responsible for the implementation of the National Water Master Plan, river basins action plans, assessments of the needs for water for all sectors in terms of quantity and quality, allocation of water resources and defining the protected areas; then this project will provide technical assistance in terms of the El Bared basin development that can then translate into policy recommendations for the Ministry. These studies having lengthy time spans so they will be done in parallel with the feasibility studies for the rehabilitation and upgrade of the existing El Bared hydro facilities and those of the solar PV farms (if found feasible). The basin development study will include, among other things, the feasibility studies for a potential pumped storage solution between the two existing dams.

Outcome 3: Policy support & Capacity Building

Output 3.1 Improved sustainable energy value chains
Output 3.2: Increased Policy and technical support for the Sustainable Energy Value Chain

The third Outcome will support improving the sustainable energy value chain which is essential to ensure that the quality of sustainable energy systems is improved both in terms of the equipment and their installation. This will be achieved through capacity building and policy support.

Output 3.1 Improved sustainable energy value chains

This activity aims also to upgrade the local capabilities throughout the complete Sustainable Energies value chain. Policy support for the standardization of the certification process for RE installations will be provided, Capacity building and trainings will be implemented for the Certification of Engineers specialized in Sustainable Energy solutions installations and throughout the installation of the renewable energy systems for each facility. In addition, workshops and policy support are envisioned within this project that would target mainly the public sector, specifically the Ministry of Energy and Water, but also the Lebanese University engineers and the hospital staff but also the Ministry of Health.
At the same time, the project will also enhance the renewable energy technologies quality value chain; that is, the various points along the chain of the renewable energy market in Lebanon where certification and/or improvement of the quality of the systems can be introduced at national, policy and

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technical levels. The details and weaknesses in the existing system will be analysed first upon project inception and detailed interventions designed accordingly. The framework approach will include the following:

- Consideration of LIBNOR standards (being the national institute that sets norms and standards for Lebanon) including creation of a set of certification criteria that will be used to assess and evaluate sustainable energy equipment, installations, and related processes. Consider factors like safety, efficiency, reliability, environmental impact, and performance and potentially also considering the procedures and systems to be put in place for the testing of renewable energy system parts with the Institute for Industrial Research Institute (IRI), being the national entity responsible for these types of interventions and to Identify any gaps or areas that need improvement to ensure they align with international best practices.
- Building the technical capacity of energy engineers working in the renewable energy sector through the advancement of a certification process in Lebanon by linking or supporting local entities to access international certification systems and curricula. The various options and types of certification and universities will be determined at a later stage. These programs should cover the necessary knowledge and skills required for designing, installing, operating, and maintaining sustainable energy systems.
- The project will also consider the existing enabling environment and consider points of entry to strengthen the certification of installed systems, based on international best practices that would define the type, size and circumstances where such certification is needed, and national structures that can accommodate such systems. This will be closely coordinated with the Lebanese Centre for Energy Conservation (LCEC).

One of the potential projects that will be considered by the project team during the inception phase is the assessment of the feasibility and complexity of implementing digitalization or data acquisition system sfor the installed renewable energy systems, including institutions like the Lebanese University and hospitals. This would entail the incorporation of advanced technologies and the establishment of a central data acquisition system to enhance the efficiency of renewable energy production and distribution. The data acquisition system will be responsible for gathering and managing data from various sources, enabling real-time monitoring, and enabling comprehensive data analysis. By way of illustration, the Ministry of Public Health (MoPH) could potentially adopt this approach to enhance energy efficiency within its hospitals, evaluate the environmental (carbon) impact, and make informed decisions regarding resource allocation. The installation of such a system is very beneficial however the capacity of the sectoral administration, whether at ministerial level or the level of EDL at utility-level, needs to be considered given that such systems would need maintenance and follow-up to remain viable and useful. They require both human and financial resources in the long-run to be allocated by the public sector.

- **Activity 3.2: Policy and Technical Support**

The first part of this component involves an in-depth exploration of the technical aspects related to the renewable energy development as the design and construction of hydroelectric facilities. Preliminary feasibility studies have been conducted by the Ministry of Energy and Water to identify potential hydroelectric sites with optimal conditions for power generation. Factors such as water availability, topography, environmental impact, and social considerations have been taken into account during the site selection process. Detailed feasibility studies will be carried out for the most promising sites to prepare the grounds for the works implementation phase.

The aim is to develop innovative and efficient designs that maximize power output while minimizing environmental impacts. Different types of hydro facilities, such as run-of-river, reservoir-based, or pumped storage, will be explored, taking into consideration local conditions and energy demands.

In addition to that, this component revolves around developing a holistic hydropower strategy for Lebanon. The hydropower strategy will be formulated to align with Lebanon's energy conservation and environmental sustainability goals. The strategy will encompass policy development, regulatory frameworks, and long-term planning for hydroelectric power generation. Emphasis will be placed on

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optimizing energy efficiency, reducing greenhouse gas emissions and developing pumped storage solutions in view of integrating and expanding the share of renewable energy into Lebanon's energy mix.

The second part of this component will address policy support to EDL and the Ministry of Energy and Water on various aspects of renewable energy generation, supply and transmission as needed to upscale renewables and disseminate lessons learnt. All stakeholders will be consulted on the subject and the Ministry of Energy and Water will be requested to assign technical focal points for the studies . UNDP will consult with the technical focal points during the preparation of the policies or technical studies and will then seek the overall endorsement of the relevant ministry before publication, depending on the type of document and its legal considerations. A preliminary assessment of the legal and administrative constraints to the development of the national hydropower market for Lebanon done in a World Bank study in 2017 (reference ref. WB study D4) has provided lots of insights on the subject and further studies will be conducted in coordination with the Ministry of Energy and Water to identify the optimal legal and administrative set-up that can foster the development of the sustainable energy market in Lebanon. Moreover, support will be provided to EDL from the contingency budget if the available funds are sufficient.

SUMMARY

Outcome 1: Increased quality and access to public services through sustainable energy solutions

Outputs Contributing to Outcome 1

1.1 Increased energy savings and access to electricity in public service facilities
- Higher Education Facilities: Lebanese University
- Public Hospitals

Activity Output 1.1

- Identification of priority sites and coordination with other partners and relevant ministries
- Technical backstopping for the design of the renewable energy and energy efficiency systems
- Energy audit reports of selected public service facilities and identification of priority interventions
- Solar photovoltaic systems interventions
- Energy lighting retrofits and control interventions
- Solar thermal systems interventions
- Heating, ventilation, and cooling interventions
- Computerized maintenance and monitoring systems interventions
- EE Interventions depending on their viability: biogas, CSP, heat pumps, heat recovery…

Outcome 2 Sustainable Energy for Electricity Supply
Outputs Contributing to Outcome 1
2.1 Increased Sustainability of Public Electricity
Activity Output 2.1

- Feasibility studies. expertise and fact findings of existing equipment and infrastructure in El Bared Hydro Facilities.
- Rehabilitation, upgrade or replacement of existing Hydro Equipment and Infrastructure at El Bared Hydro facilities.
- On the job training of the El Bared Hydro facilities personnel during the rehabilitation and upgrade works.
- Training program for the human resources, comprehensive maintenance strategy and operating procedures for an effective management of the El Bared facilities

Outcome 3: Policy support & Capacity Building

Outputs Contributing to Outcome 3
3.1 Improved sustainable energy value chains

- Increased knowledge and awareness on sustainable energy approaches

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- Improved technical capacity for beneficiaries and renewable energy suppliers on sustainable energy solutions

3.2 Increased Policy and technical support

- Improvement of the renewable energy value chain in Lebanon through the design of a certification programme with for example LIBNOR and IRI
- Advocacy in terms of renewable energy policies
-

Activity 3.1

- Certification system designed for solar system value chain including but not limited to:
  - Assessment and upgrade of IRI for testing sustainable energy equipment
  - Development of standards for sustainable energy equipment with LIBNOR
  - Curriculum development/identification and training programs for engineers.
  - Design of a certification system for certain renewable energy installations to improve quality
  - Preparation of guidelines

Activity 3.2

- Technical studies for the development of further renewable energy projects such as hydropower facilities.
- Preparation of renewable energy strategies for Lebanon such as a national hydropower strategy.
- Legal and technical policy analyses in partnership with LCEC and MoEW.
- Communication and awareness raising activities

Resources Required to Achieve the Expected Results

To ensure effectiveness of programme delivery, the action will benefit from the dedicated project team, established in a cost-effective manner, whose positions match the stated outputs of the action while maintaining UNDP's quality assurance standards, subject to possible changes as needed to accompany the evolution of national needs and priorities. Given that an EU project on renewable energy will also be under implementation at the same time, some of the human resources will also be shared as follows:

1. Project Manager (cost shared at 50%): A single project manager will oversee both projects to ensure overall alignment, coordination, and adherence to project goals and timelines. This individual will have a comprehensive understanding of the objectives of each project and will be responsible for the oversight and management of both projects.
2. Communication Associate (salary not cost-shared but will support both projects): a communication assistant will handle the preparation of communication material for both projects while ensuring alignment with UNDP's central communication team and in line with respective donor communication guidelines. This individual can ensure that messaging is consistent across both projects and maximize the impact of communication efforts.

An annual monitoring and evaluation plan, including risks and mitigating measures, will be developed based on the annual work plan designed by the project and the UNDP quality assurance team. The project is part of the overall country programme quality assurance structures, which include a solid results-based monitoring system and data collection and monitoring capacity. In addition, and based on specific punctual needs, individual consultants (experts), consulting companies and contracting companies will be procured based on UNDP procurement guidelines to fulfil and compliment specific capacity needs.

Partnerships and Stakeholders' Engagement

Public sector stakeholders are primarily the Ministry of Energy and Water and EDL but also the Ministry of Health, Ministry of Industry, Ministry of Education and Higher Education, Ministry of Environment and Ministry of Interior and Municipalities.

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Furthermore, the project will also coordinate and work closely with the Lebanese Centre for Energy Conservation (LCEC), the Association of Lebanese Industrialists (ALI), the Order of Engineers and Architects (OEA), the Lebanese Green Building Council (LGBC), the Lebanese Solar Energy Society (LSES) and other NGOs and civil society associations working in the sector of sustainable energy and climate change. These stakeholders will be engaged by the UNDP team to coordinate activities, ensure policy dialogue, exchange knowledge and information but without any financial payments.

Knowledge

The project will work extensively on capacity building and knowledge products related to sustainable energy and energy efficiency and conservation. These products will be prepared by in-house experts as well as international and local consultants and will cover various technical, policy and legislative topics in the field depending on what is most needed to push forward the climate mitigation agenda. All products such as guidelines and publications will be posted for public use on the UNDP website.

Furthermore, the project will work on knowledge building at technical level for the design, installation and operation and maintenance of renewable energy systems and energy efficiency measures particular with the engineering teams of the beneficiary sites. Targeted trainings will be conducted for these personnel to build their capacity.

Sustainability and Scaling Up

The project builds on nearly two decades of experience in the sustainable energy sector for UNDP and on existing synergies with relevant stakeholders that is on-going through the current CEDRO5 project. The project will be implemented in close coordination with the Ministry of Energy and Water as the main government focal point on energy to ensure sustainability of the Action, particularly in terms of policy support and technical work.

The project will complement the parallel project on sustainable energy funded by the EU. These two projects will work in tandem and have similar approaches but with different scopes, as follows:

- In the public sector, both projects aim to support the installation of various renewable energy and energy efficiency solutions. The EU-funded project primarily focuses on facilities such as LAF hospitals/healthcare units, LAF critical facilities, ISF barracks, courthouses, schools, etc. In contrast, the KFW project specifically targets installations at public hospitals and Lebanese universities.
- Both projects emphasize capacity building through training programs. The EU-funded project aims to provide vocational training to enhance the skills of technicians and renewable energy service providers. On the other hand, the KFW project focuses on certifying engineers with expertise in sustainable energy solution installations.
- Both projects will engage in activities related to policy support and the review and improvement of standards pertaining to the targeted technologies.

However, these projects also have unique activities that differ in their scope: the EU project includes supporting the implementation of microgrids in collaboration with the private sector, as well as continuing various initiatives initiated in Cedro 5, such as supporting startups, establishing an energy hub, and facilitating renewable energy certificates (RECs). In contrast, the KFW project introduces a new component that involves the study and upgrade of hydropower, including all necessary feasibility and technical studies.

Risks and Assumptions

Lebanon's political, economic and security situation remain unstable. This situation may last the entire duration of the project implementation period. Mitigation measures are included for each of the risks

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mentioned in the table below and will be continuously assessed and updated by UNDP. Any changing or critical risk will also be escalated to the Project Board as needed.

Table 1. Project Risks and Mitigation

Category	Risk	Likelihood (High/Medium/ Low)	Impact (High/Medium/ Low)	Mitigating measures
External environment	Damage to installed equipment and material	Medium	High	Memorandums of understanding will be signed with beneficiaries to ensure the protection, operation and maintenance of the equipment installed.
	Natural disasters and environmental degradation	Low	High	Ensure all outdoor equipment can withstand severe weather conditions expected in Lebanon. Ensure that an environmental statement is undertaken for all projects with significant installations.
Planning, processes and systems	Battery disposal	High	Medium	UNDP will be working on battery disposal through the EU-funded TaDWIR project. Activities will be coordinated with this project.
	Ability to effectively operate and maintain installed systems	High	High	A minimum of two-years performance guarantee for all systems installed will be required. The possibility of including O&M contract will also be assessed and capacity building activities will be ensured.
	Delays in import clearance will cause delays to the project	Medium	Medium	Sufficient time will be given for project implementation to account for expected delays in manufacturing, shipping and clearance of imported system components. Continuous coordination will be conducted with the contractors and suppliers to overcome any possible delays in manufacturing or shipping.
	Delays in manufacturing and shipping of equipment	High	High	
People and the organisation	Instability in the security situation in the country	Medium	Medium	Isolated security incidences may occur. Appropriate measures will be taken accordingly. These could include the postponement of some of the works on site, and the coordination with the Internal Security Forces and UN Department of Security as needed.

Visibility

Communication will be an integral part of the new phase of this intervention whereby activities and messages will support project implementation and enhance visibility, at the national and local levels, of all different projects and activities and main stakeholders including the German Government, BMZ, and KfW. The joint UNDP-KfW communication guidelines will be followed on all communication and visibility material and for all events, audio-visual productions and reports.

The main target group will be segmented as follows:

- Media: the results of the project will be mainly communicated through social media networks.
- Local communities: UNDP will reach out to the various villages and communitieswhere implementation is taking place.. The project will ensure donor visibility throughout the process in terms of signage on site and information sharing with beneficiaries. UNDP will set up project sign boards that will contain at least the following message:

"A development project of @, co-financed by the Federal Republic of Germany through KfW Development Bank." A project seal provided by the German Embassy shall be placed on the project sign board.

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The main visibility activities will be press releases (if and when applicable), social media posts, videos, success stories, meetings with partners, partnership events. KfW will be tagged in the social media posts including KfW projects (Instagram: KfW stories, Twitter: @KfW_Beirut). UNDP will provide KfW with photos and videos within the framework of the quarterly reports.

UNDP will submit proposals for inauguration, launching and closing events, exhibitions etc. to KfW. Press releases will be issued and cover the following but not limited events:

- Event upon project completion (at site level)
- Trainings / workshops for various stakeholders
- Closing event (1 press release)

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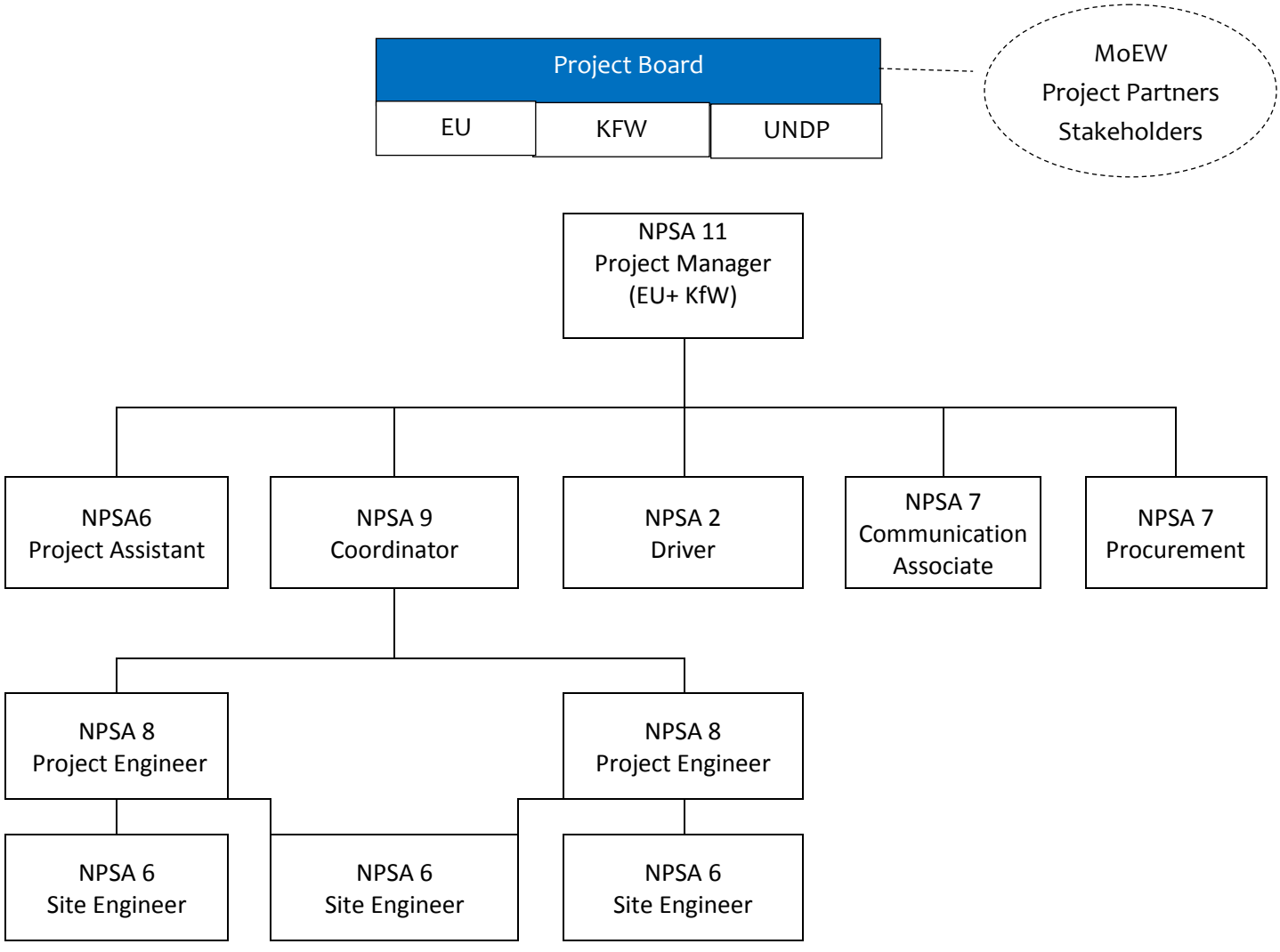
IV. PROJECT MANAGEMENT

The existing UNDP Project Manager and the team will be responsible for the day-to-day management of the project. The implementations of the envisaged projects in renewable energy and energy efficiency will be subcontracted to private construction companies through UNDP's competitive procurement processes. The implementation of works on the ground will also closely liaise with the on-going work with municipalities and the Ministry of Energy and Water and EDL. Coordination with other UNDP projects that are under implementation such as the Lebanon Host-Communities Project (LHSP) will also take place.

The project is expected to start in 2024 and is fully harmonised with the EU-funded CEDRO6 project which follows a similar objective and activities. To ensure efficiency and coordination, UNDP will endeavour to align the design, implementation modality, and management structure of this project with the EU-funded CEDRO6 project. As both projects are developed in more detail during 2023, all partners will define several aspects of their management, including reporting requirements, project board structure and joint coordination meetings, project management units of UNDP, and joint communication, with the aim of aligning the following to the greatest extent possible.

- Reporting requirements
- Project board structure and joint coordination meetings
- Project management units of UNDP, including the sharing of personnel and operating costs to ensure efficiency and cost-savings
- Joint communication

The project board and project management structure will be as per below:



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V. RESULTS FRAMEWORK

Intended Outcome as stated in the UNSDCF/Country Programme Results and Resource Framework:
3.1. Environmental Governance Improved.

Outcome indicators as stated in the Country Programme [or Regional] Results and Resources Framework, including baseline and targets:
CPD Outcome 3: Strengthened stabilization and green recovery to reduce vulnerabilities and environmental risks, including through enhanced competitiveness and business environment of sustainability-oriented MSMEs and high potential green productive sector values chains.
CPD output 3.2 (IRRF output 5.2): Transition to renewable energy accelerated capitalizing on technological gains, clean energy innovations and new financing mechanisms to support green recovery.
CPD output indicator 3.2.1.: Amount of renewable energy installed annually (output indicator for UNSDCF output 1.2, IRRF 5.2.1. Increase (in megawatt) in installed renewable energy capacity with solar, wind, biomass and other technologies)

Applicable Output(s) from the UNDP Strategic Plan:
1.5.1. Solutions adopted to achieve universal access to clean, affordable and sustainable energy
2.51. Solutions developed, financed and applied at scale for energy efficiency transformation to clean energy and zero carbon development for poverty eradication and structural transformation

Project title and Quantum Project Number: Improving energy security and promoting the transition to sustainable energy in Lebanon - Project Number: 01001708

EXPECTED OUTPUTS (Quantum structure)	OUTPUT INDICATORS	DATA SOURCE	BASELINE		TARGETS (by frequency of data collection)						DATA COLLECTION METHODS & RISKS
			Value	Year	Year 1	Year 2	Year 3	Year 4	Year 5	FINAL	
Output 1: Improving energy security and promoting the transition to sustainable energy in Lebanon											
Activity 1.1. Increased energy savings & access to electricity from small scale renewable energy installations in public service facilities	**1.1.1.** Tons of CO2 emissions saved per year with KfW support	UNDP, MoEW, MoE, Beneficiries	TBD	2023	275	1,975	3,675	5,100	5,670	**5,670**	Report KfW-funded intervention's M&E system, Progress reports, reviews, evaluations
	1.1.2. Energy saved per year (MWh) with EU Support		TBD	2023	450	3,000	5,550	7,700	8,550	**8,550**	Baseline and endline surveys
	1.1.3. Renewable energy power generated per year (MWh) with KfW Support		TBD	2023	280	2,310	4,340	6,090	6,770	**6,770**	Baseline and endline surveys
	1.1.4 Estimated economic savings (LCOE 0.27 USD/KWh)		TBD	2023	$75,600	$623,700	$1,171,800	$1,644,300	$1,827,900	**$1,827,900**	Baseline and endline surveys
	1.1.5 Number of beneficiaries with increased access to energy services		TBD	2023	22,500	73,500	124,500	153,000	170,000	**170,000**	Baseline and endline surveys
Activity 2.1. Increased sustainable	2.1.1 Renewable energy power generated per year (MWh)	UNDP, MSME	TBD	2023	0	0	0	0	47,143	47,143	Report KfW-funded intervention's M&E system, Progress reports,

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

EXPECTED OUTPUTS (Quantum structure)	OUTPUT INDICATORS	DATA SOURCE	BASELINE		TARGETS (by frequency of data collection)						DATA COLLECTION METHODS & RISKS
			Value	Year	Year 1	Year 2	Year 3	Year 4	Year 5	FINAL	
energy sources for electricity supply		s,									reviews, evaluations
	2.1.2 Tons of CO2 emissions saved per year	Beneficiaries	TBD	2023	0	0	0	0	31,000	31,000	Report EU-funded intervention's M&E system, Progress reports, reviews, evaluations
	2.1.3 Number of beneficiaries with increased access to energy services		*TBD*	2023	0	0	0	0	3,600	3,600	Baseline and endline surveys
Activity 3.1. Improved solar energy value chains	3.1.1 Certification of installations/projects		TBD	2023	0	1	3	4	4	4	Baseline and endline surveys
	3.1.2 Number of engineers trained for certification		TBD	2023	0	50	100	150	150	150	Baseline and endline surveys
	3.1.3 Number of sustainable energy equipment types validation enabled and/or enhanced		TBD	2023	0	1	2	3	3	3	Baseline and endline surveys
	3.1.4. Number of sustainable energy standards reviewed and updated		TBD	2023	2	6	10	12	12	12	Baseline and endline surveys
Activity 3.2 Increased policy and technical support for the sustainable energy value chain	3.2.1 Number of sustainable energy related studies conducted		TBD	2023	0	1	3	4	5	5	Baseline and endline surveys
	3.2.2 Number of legal and policy support initiatives conducted		TBD	2023	0	1	2	3	3	3	Baseline and endline surveys

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

VI. MONITORING AND EVALUATION

In accordance with UNDP's programming policies and procedures the project will be monitored through the following monitoring and evaluation plans

Monitoring Plan

Monitoring Activity	Purpose	Frequency	Expected Action	Partners (If joint)	Cost (If any)
Track results progress	Progress data against the results indicators in the RRF will be collected and analysed to assess the progress of the project in achieving the agreed outputs.	Quarterly, or in the frequency required for each indicator.	Slower than expected progress will be addressed by project management.	UNDP	Internal
Monitor and Manage Risk	Identify specific risks that may threaten achievement of intended results. Identify and monitor risk management actions using a risk log. This includes monitoring measures and plans that may have been required as per UNDP's Social and Environmental Standards. Audits will be conducted in accordance with UNDP's audit policy to manage financial risk.	Quarterly	Risks are identified by project management and actions are taken to manage risk. The risk log is actively maintained to keep track of identified risks and actions taken.	UNDP	Internal
Learn	Knowledge, good practices and lessons will be captured regularly, as well as actively sourced from other projects and partners and integrated back into the project.	At least annually	Relevant lessons are captured by the project team and used to inform management decisions.	UNDP	Internal
Project Quality Assurance	The quality of the project will be assessed against UNDP's quality standards to identify project strengths and weaknesses and to inform management decision making to improve the project.	Every two years	Areas of strength and weakness will be reviewed by project management and used to inform decisions to improve project performance.	UNDP	Internal
Review and Make Course Corrections	Internal review of data and evidence from all monitoring actions to inform decision making.	At least annually	Performance data, risks, lessons and quality will be discussed by the project board and used to make course corrections. UNDP will also undertake prior consultation with KfW	UNDP	Internal
Project Progress Report	A summary of Project Progress Report will be presented to the Project Board and key stakeholders, consisting of progress data showing the results achieved against pre-defined annual targets at the output level.	Semi-Annual, and at the end of the project (final report)	Assessing status of project implementation and flagging any updated risks or potential changes/need for adaptive management to the project scope.	UNDP	Internal
Project Review (Project Board)	The project's governance mechanism (i.e., Project Board) will hold regular project reviews to assess the	Annually	Any quality concerns or slower than expected progress should be discussed by the Project	UNDP	Internal

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	performance of the project and review the Multi-Year Work Plan to ensure realistic budgeting over the life of the project. In the final year, the Project Board shall hold an end-of project review to capture lessons learned and discuss opportunities for scaling up and to socialize project results and lessons learned with relevant audiences.		Board and management actions agreed to address the issues identified.		

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VII. MULTI-YEAR WORK PLAN

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EXPECTED OUTPUTS	PLANNED ACTIVITIES	Planned Budget by Year					RESPONSIBLE PARTY	PLANNED BUDGET		
		Y1 - 2024	Y2 - 2025	Y3 - 2026	Y4 - 2027	Y5 - 2028		Funding Source	Budget Description	Amount
Output 1: Improving energy security and promoting the transition to sustainable energy in Lebanon Gender marker: 2	**1.1. Increased energy savings & access to electricity from small scale renewable energy installations in public service facilities**	32,023.30	32,023.30	32,023.30	32,023.30	32,023.30	UNDP	KfW	64300-Staff Mgmt Costs - IP Staff	160,116.50
		122,203.40	122,203.40	125,035.40	125,035.40	125,035.40			71400-Contractual Services - Individ	619,513.00
		619,201.73	2,234,049.49	2,234,049.49	1,690,360.17	795,024.77			72100-Contractual Services-Companies	7,572,685.65
		11,327.00	11,327.00	11,327.00	11,327.00	11,327.00			73100-Rental & Maintenance-Premises	56,635.00
		2,832.00	2,832.00	-	-	-			72800-Information Technology Equipmt	5,664.00
		5,394.00	5,394.00	5,394.00	5,394.00	5,394.00			71600-Travel	26,970.00
	Public University RE & EE Interventions, digitalisation	1,348.00	1,348.00	1,348.00	1,348.00	1,348.00			72500-Supplies	6,740.00
	Public Hospitals RE & EE Interventions, digitalisation	1,888.00	1,888.00	1,888.00	1,888.00	1,888.00			72300-Materials & Goods	9,440.00
	Technical backstopping	215.80	215.80	215.80	215.80	215.80			74500-Miscellaneous Expenses	1,079.00
		5,663.00	5,663.00	5,663.00	5,663.00	5,663.00			72200-Equipment and Furniture	28,315.00
		64,167.70	193,355.52	193,355.52	149,860.37	78,233.54			75100-Facilities & Administration	678,972.65
		866,263.93	**2,610,299.51**	**2,610,299.51**	**2,023,115.04**	**1,056,152.81**				**9,166,130.80**
	2.1. Increased sustainable energy sources for electricity supply	32,023.30	32,023.30	32,023.30	32,023.30	32,023.30	UNDP	KfW	64300-Staff Mgmt Costs - IP Staff	160,116.50
		122,203.40	122,203.40	125,035.40	125,035.40	125,035.40			71400-Contractual Services - Individ	619,513.00
	Design & supervision for Nahr El Bared hydro rehab + environmental studies	43,149.95	258,899.68	954,404.10	1,779,358.36	1,736,208.41			72100-Contractual Services-Companies	4,772,020.50
	Feasibility study for Nahr El Bared solar farm dev & pumped storage	11,327.00	11,327.00	11,327.00	11,327.00	11,327.00			73100-Rental & Maintenance-Premises	56,635.00
	Nahr El Bared hydro facility rehabilitation/upgrade works	2,832.00	2,832.00	-	-	-			72800-Information Technology Equipmt	5,664.00
		5,394.00	5,394.00	5,394.00	5,394.00	5,394.00			71600-Travel	26,970.00
		1,348.00	1,348.00	1,348.00	1,348.00	1,348.00			72500-Supplies	6,740.00
		1,888.00	1,888.00	1,888.00	1,888.00	1,888.00			72300-Materials & Goods	9,440.00
		215.80	215.80	215.80	215.80	215.80			74500-Miscellaneous Expenses	1,079.00
		5,663.00	5,663.00	5,663.00	5,663.00	5,663.00			72200-Equipment and Furniture	28,315.00
		18,083.56	35,343.53	90,983.89	156,980.23	153,528.23			75100-Facilities & Administration	454,919.44
		244,128.01	**477,137.71**	**1,228,282.49**	**2,119,233.09**	**2,072,631.14**				**6,141,412.44**
	3.1. Improved solar energy value chains	32,023.30	32,023.30	32,023.30	32,023.30	32,023.30	UNDP	KfW	64300-Staff Mgmt Costs - IP Staff	160,116.50
		122,203.40	122,203.40	125,035.40	125,035.40	125,035.40			71400-Contractual Services - Individ	619,513.00
	Sustainable energy testing equipment	15,533.98	79,611.65	432,901.83	454,153.18	36,138.08			72100-Contractual Services-Companies	1,018,338.72
	Sustainable energy standards	11,327.00	11,327.00	11,327.00	11,327.00	11,327.00			73100-Rental & Maintenance-Premises	56,635.00
	Certification for Sustainable Energy Engineers	2,832.00	2,832.00	-	-	-			72800-Information Technology Equipmt	5,664.00
	Workshops	5,394.00	5,394.00	5,394.00	5,394.00	5,394.00			71600-Travel	26,970.00
	Communication	1,348.00	1,348.00	1,348.00	1,348.00	1,348.00			72500-Supplies	6,740.00
		1,888.00	1,888.00	1,888.00	1,888.00	1,888.00			72300-Materials & Goods	9,440.00
		215.80	215.80	215.80	215.80	215.80			74500-Miscellaneous Expenses	1,079.00
		5,663.00	5,663.00	5,663.00	5,663.00	5,663.00			72200-Equipment and Furniture	28,315.00
		15,874.28	21,000.49	49,263.71	50,963.81	17,522.61			75100-Facilities & Administration	154,624.90
		214,302.76	**283,506.64**	**665,060.04**	**688,011.49**	**236,555.19**				**2,087,436.12**
	3.2. Increased policy and technical support for the sustainable energy value chain	32,023.30	32,023.30	32,023.30	32,023.30	32,023.30	UNDP	KfW	64300-Staff Mgmt Costs - IP Staff	160,116.50
		122,203.40	122,203.40	125,035.40	125,035.40	125,035.40			71400-Contractual Services - Individ	619,513.00
	Technical studies + hydropower strategy	-	368,932.04	368,932.04	245,954.69	245,954.69			72100-Contractual Services-Companies	1,229,773.46
	Legal and Policy support initiatives	11,327.00	11,327.00	11,327.00	11,327.00	11,327.00			73100-Rental & Maintenance-Premises	56,635.00
		2,832.00	2,832.00	-	-	-			72800-Information Technology Equipmt	5,664.00
		5,394.00	5,394.00	5,394.00	5,394.00	5,394.00			71600-Travel	26,970.00
		1,348.00	1,348.00	1,348.00	1,348.00	1,348.00			72500-Supplies	6,740.00
		1,888.00	1,888.00	1,888.00	1,888.00	1,888.00			72300-Materials & Goods	9,440.00
		215.80	215.80	215.80	215.80	215.79			74500-Miscellaneous Expenses	1,078.99
		5,663.00	5,663.00	5,663.00	5,663.00	5,663.00			72200-Equipment and Furniture	28,315.00
			107,876.90	*215,749.73*	*107,874.87*	*107,874.87*			*74500-Miscellaneous Expenses*	539,376.37
		14,631.56	52,776.28	61,406.10	42,937.92	42,937.92			75100-Facilities & Administration	214,689.79
		197,526.06	**712,479.72**	**828,982.37**	**579,661.98**	**579,661.97**				**2,898,312.11**
TOTAL	**Total**	**1,522,220.75**	**4,083,423.58**	**5,332,624.41**	**5,410,021.61**	**3,945,001.11**				**20,293,291.47**

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VIII. GOVERNANCE AND MANAGEMENT ARRANGEMENTS

The Project will be executed under the UNDP Direct Implementation Modality (DIM), whereby UNDP will act as the executing and implementing agency. The UNDP will monitor the progress towards intended results, and will ensure high-quality managerial, technical and financial implementation of the project, and will be responsible for monitoring and ensuring proper use of administrated funds to the assigned activities, timely reporting of implementation progress as well as undertaking of mandatory and non-mandatory evaluations for each of their respective components. Furthermore, the procurement of goods and services and the recruitment of personnel shall be provided in accordance with UNDP guidelines, procedures and regulations.

The UNDP country office will provide the following support services, covered by Direct Project Cost (DPC), for the activities of the programme/project:

i. Payments, disbursements and other financial transactions

ii. Recruitment of staff, project personnel, and consultants

iii. Procurement of services and equipment, including disposal

iv. Organization of training activities, conferences, and workshops, including fellowships

v. Travel including visa requests, ticketing, and travel arrangements

vi. Shipment, customs clearance, vehicle registration, and accreditation

vii. Security management service and Malicious Acts Insurance Policy

viii. Access to Quantum for project managers and other staff, Payroll management services and Medical Clearance Services for all staff

The UNDP country office will also provide the following general oversight and management services at the rate of 8%, for the activities of the project which include the following:

i. Corporate executive management and resource mobilisation

ii. Corporate accounting, financial management, internal audit, legal support and human resources management

iii. Policy guidance and Bureau/Country Office management

The audit of DIM projects is made through the regular external (UN Board of Auditors) or internal audits (audits managed by UNDP's Office of Audit and Investigations).

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IX. LEGAL CONTEXT

This project document shall be the instrument referred to as such in Article 1 of the Standard Basic Assistance Agreement between the Government of Lebanon and UNDP, signed in 1986. All references in the SBAA to "Executing Agency" shall be deemed to refer to "Implementing Partner."

This project will be implemented by UNDP in accordance with its financial regulations, rules, practices and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. Where the financial governance of an Implementing Partner does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition, the financial governance of UNDP shall apply.

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

X. RISK MANAGEMENT

1. UNDP as the Implementing Partner will comply with the policies, procedures and practices of the United Nations Security Management System (UNSMS.)

2. UNDP as the Implementing Partner will undertake all reasonable efforts to ensure that none of the [project funds][12] are used to provide support to individuals or entities associated with terrorism , that the recipients of any amounts provided by UNDP hereunder do not appear on the United Nations Security Council Consolidated Sanctions List, and that no UNDP funds received pursuant to the Project Document are used for money laundering activities. The United Nations Security Council Consolidated Sanctions List can be accessed via https://www.un.org/securitycouncil/content/un-sc-consolidated-list. This provision must be included in all sub-contracts or sub-agreements entered into under this Project Document.

3. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).

4. UNDP as the Implementing Partner will: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.

5. In the implementation of the activities under this Project Document, UNDP as the Implementing Partner will handle any sexual exploitation and abuse ("SEA") and sexual harassment ("SH") allegations in accordance with its regulations, rules, policies and procedures.

6. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

7. UNDP as the Implementing Partner will ensure that the following obligations are binding on each responsible party, subcontractor, and sub-recipient:

a. Consistent with the Article III of the SBAA *[or the Supplemental Provisions to the Project Document]*, the responsibility for the safety and security of each responsible party, subcontractor and sub-recipient and its personnel and property, and of UNDP's property in such responsible party's, subcontractor's and sub-recipient's custody, rests with such responsible party, subcontractor and sub-recipient. To this end, each responsible party, subcontractor and sub-recipient shall:
   i. put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;
   ii. assume all risks and liabilities related to such responsible party's, subcontractor's and sub-recipient's security, and the full implementation of the security plan.

b. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the responsible party's, subcontractor's and sub-recipient's obligations under this Project Document.

c. Each responsible party, subcontractor and sub-recipient (each a "sub-party" and together "sub-parties") acknowledges and agrees that UNDP will not tolerate sexual harassment and sexual exploitation and abuse of anyone by the sub-parties, and other entities involved in Project implementation, either as

[12] To be used where UNDP is the Implementing Partner

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

contractors or subcontractors and their personnel, and any individuals performing services for them under the Project Document.

(a) In the implementation of the activities under this Project Document, each sub-party shall comply with the standards of conduct set forth in the Secretary General's Bulletin ST/SGB/2003/13 of 9 October 2003, concerning "Special measures for protection from sexual exploitation and sexual abuse" ("SEA").

(b) Moreover, and without limitation to the application of other regulations, rules, policies and procedures bearing upon the performance of the activities under this Project Document, in the implementation of activities, each sub-party, shall not engage in any form of sexual harassment ("SH"). SH is defined as any unwelcome conduct of a sexual nature that might reasonably be expected or be perceived to cause offense or humiliation, when such conduct interferes with work, is made a condition of employment or creates an intimidating, hostile or offensive work environment. SH may occur in the workplace or in connection with work. While typically involving a pattern of conduct, SH may take the form of a single incident. In assessing the reasonableness of expectations or perceptions, the perspective of the person who is the target of the conduct shall be considered.

d. In the performance of the activities under this Project Document, each sub-party shall (with respect to its own activities), and shall require from its sub-parties (with respect to their activities) that they, have minimum standards and procedures in place, or a plan to develop and/or improve such standards and procedures in order to be able to take effective preventive and investigative action. These should include: policies on sexual harassment and sexual exploitation and abuse; policies on whistleblowing/protection against retaliation; and complaints, disciplinary and investigative mechanisms. In line with this, sub-parties will and will require that their respective sub-parties will take all appropriate measures to:

   (i) Prevent its employees, agents or any other persons engaged to perform any services under this Project Document, from engaging in SH or SEA;

   (ii) Offer employees and associated personnel training on prevention and response to SH and SEA, where sub-parties have not put in place its own training regarding the prevention of SH and SEA, sub-parties may use the training material available at UNDP;

   (iii) Report and monitor allegations of SH and SEA of which any of the sub-parties have been informed or have otherwise become aware, and status thereof;

   (iv) Refer victims/survivors of SH and SEA to safe and confidential victim assistance; and

   (v) Promptly and confidentially record and investigate any allegations credible enough to warrant an investigation of SH or SEA. Each sub-party shall advise UNDP of any such allegations received and investigations being conducted by itself or any of its sub-parties with respect to their activities under the Project Document, and shall keep UNDP informed during the investigation by it or any of such sub-parties, to the extent that such notification (i) does not jeopardize the conduct of the investigation, including but not limited to the safety or security of persons, and/or (ii) is not in contravention of any laws applicable to it. Following the investigation, the relevant sub-party shall advise UNDP of any actions taken by it or any of the other entities further to the investigation.

e. Each sub-party shall establish that it has complied with the foregoing, to the satisfaction of UNDP, when requested by UNDP or any party acting on its behalf to provide such confirmation. Failure of the relevant sub-party to comply of the foregoing, as determined by UNDP, shall be considered grounds for suspension or termination of the Project.

f. Each responsible party, subcontractor and sub-recipient will ensure that any project activities undertaken by them will be implemented in a manner consistent with the UNDP Social and Environmental Standards and shall ensure that any incidents or issues of non-compliance shall be reported to UNDP in accordance with UNDP Social and Environmental Standards.

g. Each responsible party, subcontractor and sub-recipient will take appropriate steps to prevent misuse of funds, fraud, corruption or other financial irregularities, by its officials, consultants, subcontractors and sub-recipients in implementing the project or programme or using the UNDP funds. It will ensure that its financial management, anti-corruption, anti-fraud and anti money laundering and countering the financing of terrorism policies are in place and enforced for all funding received from or through UNDP.

h. The requirements of the following documents, then in force at the time of signature of the Project Document, apply to each responsible party, subcontractor and sub-recipient: (a) UNDP Policy on Fraud and other Corrupt Practices (b) UNDP Anti-Money Laundering and Countering the Financing of Terrorism

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Policy; and (c) UNDP Office of Audit and Investigations Investigation Guidelines. Each responsible party, subcontractor and sub-recipient agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.

i. In the event that an investigation is required, UNDP will conduct investigations relating to any aspect of UNDP programmes and projects. Each responsible party, subcontractor and sub-recipient will provide its full cooperation, including making available personnel, relevant documentation, and granting access to its (and its consultants', subcontractors' and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with it to find a solution.

j. Each responsible party, subcontractor and sub-recipient will promptly inform UNDP as the Implementing Partner in case of any incidence of inappropriate use of funds, or credible allegation of fraud, corruption other financial irregularities with due confidentiality.

   Where it becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, each responsible party, subcontractor and sub-recipient will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit and Investigations (OAI). It will provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

k. UNDP will be entitled to a refund from the responsible party, subcontractor or sub-recipient of any funds provided that have been used inappropriately, including through fraud corruption or other financial irregularities, or otherwise paid other than in accordance with the terms and conditions of the Project Document. Such amount may be deducted by UNDP from any payment due to the responsible party, subcontractor or sub-recipient under this or any other agreement.

   Where such funds have not been refunded to UNDP, the responsible party, subcontractor or sub-recipient agrees that donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities under this Project Document, may seek recourse to such responsible party, subcontractor or sub-recipient for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud, corruption or other financial irregularities, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

   Note: The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with responsible parties, subcontractors and sub-recipients.

l. Each contract issued by the responsible party, subcontractor or sub-recipient in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal, have been given, received, or promised in connection with the selection process or in contract execution, and that the recipient of funds from it shall cooperate with any and all investigations and post-payment audits.

m. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project or programme, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover and return any recovered funds to UNDP.

n. Each responsible party, subcontractor and sub-recipient shall ensure that all of its obligations set forth under this section entitled "Risk Management" are passed on to its subcontractors and sub-recipients and that all the clauses under this section entitled "Risk Management Standard Clauses" are adequately reflected, *mutatis mutandis*, in all its sub-contracts or sub-agreements entered into further to this Project Document.

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

XI. ANNEXES

1. **Project Quality Assurance Report:** Completed in Quantum/Unity+ – ref: ASM-03068
2. **Social and Environmental Screening:** Enclosed and done in System – ref: ASM-03069
3. **Risk Analysis**
4. **Capacity Assessment:** Not applicable as all activities are directly implemented by UNDP
5. **On-Granting Provisions Applicable to the Implementing Partner.** Not Applicable

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

Annex 2 Social and Environmental Screening Template

UNDP SOCIAL AND ENVIRONMENTAL SCREENING TEMPLATE (V. JULY 2022)

The completed template, which constitutes the Social and Environmental Screening Report, must be included as an annex to the Project Document at the design stage. Note: this template will be converted into an online tool. The online version will guide users through the process and will embed relevant guidance.

Project Information

Project Information	
1. Project Title	Improving energy security and promoting the transition to sustainable energy in Lebanon
2. Project Number (i.e. Quantum project ID, PIMS+)	Project Number: 01001708
3. Location (Global/Region/Country)	Lebanon
4. Project stage (Design or Implementation)	Design
5. Date	May 2023

Part A. Integrating Programming Principles to Strengthen Social and Environmental Sustainability

QUESTION 1: How Does the Project Integrate the Programming Principles in Order to Strengthen Social and Environmental Sustainability?
Briefly describe in the space below how the project mainstreams the human rights-based approach
The project strategy consists of approaching the transition to renewable energy in Lebanon at three levels: one is to provide and improve access to the public sector, and the second is to increase sustainable energy sources through rehabilitation of the existing hydropower plants, and the third is to support the renewable energy value chain and promote innovation to improve solar energy supply to greening recovery and creating jobs. Parallel to these three approaches, improving capacities, policies and legislation to push the sector forward will also be tackled based on the lessons learnt and needs identified during implementation at the field level of both components. Thus, the project seeks to further the realization of human rights by improving the access and availability of energy for all populations in Lebanon, which is fundamental to any human rights. There are no potential adverse impacts of this project that might affect human rights as the project will result in the further promotion of renewable energy solutions and the reduction of environmental degradation and its negative impacts on public health.
Briefly describe in the space below how the project is likely to improve gender equality and women's empowerment
The project recognizes the vital role of women in the relevant sector and all activities and approaches will be gender-sensitive and data collected will include gender markers to the extent possible and available. Gender-balanced representation in the national dialog, deliberate inclusion of electricity end-uses prioritized by community women at the

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feasibility stage, and outreach and training support to women, youth, and the financial institutions that serve them are examples of actions meant to prevent unbalanced appropriation by men of solar PV, services, and benefits.

In addition, the project aims to improve gender equality in the field of renewable energy by specifically targeting women in the training component to ensure that more women are empowered to work in this sector. Also, gender issues will be targeted in the work that is planned in the energy hub to promote the recruitment of women engineers in this field given that, to date, the statistics are low. Furthermore, all project activities will be mindful of gender participation.

Briefly describe in the space below how the project mainstreams sustainability and resilience

Environmental benefits of the renewable energy systems to be installed target mostly the reduction of pollution from self-generation diesel use, mainly carbon dioxide (a green-house gas – GHG), carbon monoxide, particulate matter, nitrogen oxides, and sulphur dioxide among other. They also will target delivering cleaner energy in terms of kWh than that of the Lebanese grid, which is mostly composed of fuel oil-generated power. In some occasions, the technologies selected will enable the complete shutting down of diesel generators, reducing thus the level of noise pollution as well. Therefore, the project not only targets climate mitigation action but also has environmental sustainability benefits and promotes the integration of SDG goals such as gender and sustainable cities as well.

Briefly describe in the space below how the project strengthens accountability to stakeholders

The project design will involve close interaction and participation of stakeholders mainly, government, municipalities, and the general public etc. The project will raise awareness among project stakeholders and create channels for communication through website, social media, telephone and government and project offices. The project will also disseminate information through online and printed media for relevant stakeholders to access in all languages to ensure that everyone is updated on project developments. The project website will also accommodate grievances and a redress mechanism will also be put in place. In addition, UNDP's Accountability Mechanism, which includes the Social and Environmental Compliance Review (SECU) and Stakeholder Response Mechanism (SRM) will also serve as an additional layer of grievance redress and empower stakeholders to push for accountability.

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Part B. Identifying and Managing Social and Environmental Risks

QUESTION 2: What are the Potential Social and Environmental Risks? *Note: Complete SESP Attachment 1 before responding to Question 2.*	**QUESTION 3: What is the level of significance of the potential social and environmental risks?** *Note: Respond to Questions 4 and 5below before proceeding to Question 5*			**QUESTION 6: Describe the assessment and management measures for each risk rated Moderate, Substantial or High**
Risk Description (broken down by event, cause, impact)	***Impact and Likelihood (1-5)***	***Significance (Low, Moderate Substantial, High)***	***Comments (optional)***	***Description of assessment and management measures for risks rated as Moderate, Substantial or High***
Risk 1: Duty bearers and other relevant stakeholders such as municipalities and public institutions may fall short of capacities to meet their environmental obligations and compliances. Related to: • Human Rights; P.2, P.3 • Accountability; P.14	I = 2 L = 2	Low	Due to the multiple crises, the capacity of local municipalities and the government has been drastically worsened in Lebanon. Can you please elaborate more on the EDL capacity EDL capacity is currently very limited and rapidly being depleted with an average of 100 resignations per month, The current level of salaries being still in LBP it's equivalent value in USD is very low and most people are leaving. EDL is also suffering from cash flow problems as the collected bills are not sufficient to cover for the fuel bills despite the recent increase in tariff. Thereby EDL is having an increasing difficulty in providing the services.	The objective of this project is to mitigate those negative impacts of the crises by improving the energy security of Lebanese public sector institutions. This will ultimately improve the physical living conditions of the most vulnerable people. In addition, the project will provide capacity building will for the facility staff of the targeted public institutions so that they can maintain the system properly after the intervention.
Risk 2: Marginalization of vulnerable groups when selecting the sites for environmental projects and installation of solar PV systems Related to:	I = 2 L = 2	Low	Site selection for installation of solar PV may marginalize stakeholders from participating in this sector, or from having equal level of access to energy if selection is not done in	The target groups of the project are mainly the more vulnerable people that access the public healthcare sector and the public university. The hydropower rehabilitation project is also located in one of the most vulnerable areas in north Lebanon.

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• Human Rights; P.5, P.6 • Accountability; P.13, P.14			collaboration with all stakeholders. On the other hand, the project will focus on public institution-level support (e.g. public university, hospitals) based on the vulnerability assessment to benefit the targeted community as broadly as possible. In such intervention, marginalization of specific group is much less likely although the clear communication is still needed. Therefore, the significance of this risk is rated at 'low'.	The project GRM will also help in managing this risk by providing a means for affected stakeholders to raise concerns and/or grievances.
Risk 3: Reproducing existing discriminations against women through excluding them from decision-making on project activities, benefiting from project outputs and capacity building initiatives Related to: • Gender Equality and Women Empowerment; P.10	I = 3 L = 3	Moderate	The male dominated nature of the energy sector and the limited social status and opportunities identified for women. UNDP's study identified lack of STEM education awareness for women as one of the challenges in Lebanon (2016). According to the study, women participation in various social, economic and political activities remained low with a female participation of "23.5% compared to 70.3% for men". This may pose a challenge to ensure that women will have the chance to participate at the decisions-making level.	Gender-balanced representation in the national dialog, deliberate inclusion of electricity end-uses prioritized by community women at the feasibility stage, and outreach and training support to women, youth, and the financial institutions that serve them are examples of actions meant to prevent unbalanced appropriation by men of solar PV technology, services, and benefits.
Risk 4: Climate events and disasters (including floods) on the project sites Related to:	I = 2 L = 2	Low	Extreme weather events due to climate change might affect infrastructure in Lebanon. Intense precipitation amounts can lead to the flooding of infrastructure. All solar PV	UNDP undertook the flood risk management and mapping fpr Lebanon (2016). Based on this assessment, the project will assess the flood risk for the potential sites to minimize the risk exposure. This relates to the Nahr El Bared hydropower site in particular.

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

• Standard 2: Climate Change and Disaster Risks; 2.2 • Standard 3: Community Health, Safety and Security; 3.3			systems are by-nature open air structures exposed to climate events and involve built structures that may be vulnerable to the impacts of such an event. Due to the fact that the sites have not yet been selected, the likelihood of this risk is not known, and a conservative approach is adopted. Can you please elaborate on the intervention on the Hydropower plant. Does Climate and other risks influence its work. Floods? The Hydropower plants are only being rehabilitated thereby there is no new risk involved.	
Risk 13: Working conditions not in line with national and international standards (by contractor or other entities involved in the project) Related to: • Standard 7: Labor and Working Conditions; 7.1, 7.2, 7.3, 7.4, 7.5, 7.6	I = 3 L = 2	Moderate	Installation of solar PV, rehabilitation of the HPPand/or other environmental approaches will require labour, some of which may be sourced to unskilled/manual labourers who could be less familiar with the type of installations considered for this project and the concomitant occupational health and safety requirements and risks. In addition, manufacturers, suppliers, subcontractors and subcontractors within the solar supply chain may not be in line with SES and thus procurement of solar PV may contribute to working conditions that undermine worker human rights, health and safety.	For each procurement, labour standards, social safeguards and an Occupational Safety and Health (OSH) measures will be applied for the project to ensure labour standards and rights are upheld for project workers. The Contractors will be closely monitored by the UNDP during the project implementation for their adherence to these standards.

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

Risk 6: Generation of hazardous waste (specifically e-waste) from the installed solar PV Related to: • Standard 8: Pollution Prevention and Resource Efficiency; 8.2	I = 2 L = 4	Moderate	Very modest amounts of waste will be generated during installation of solar PV (ground movement and concrete residues); electric wiring and insulator ends; broken or rejected parts and components. At the end of its lifespan, electronic waste ("e-waste") in the form of solar panels and/or batteries will be generated. Without proper handling directives, disposal and/or recycling mandate for obsolete equipment, this could result in additional waste generation, including of hazardous/phase-outs materials, chemicals or other pollutants (e.g. from batteries). Failure to recycle non-hazardous waste could also contribute to additional waste generation.	Lebanon has no functioning formal waste management infrastructure for e-waste and batteries. To address this issue, the UNDP is now implementing the EU-funded project 'Decentralised Waste Management Integrated Response (2021-2024)'. The project aims at reaching all of the sorted material to be collected and a maximum recycling of E-waste and batteries through a system of EPR, preferably to be implemented voluntarily by the sector within constraints set by the national government and in close cooperation with the municipalities. Therefore, the above UNDP's project will serve as mitigation measures for this project.

QUESTION 4: What is the overall project risk categorization?		
Low Risk	☐	
Moderate Risk	**X**	Six potential risks have been identified for this project, three of which are assessed as MODERATE and three as LOW. As a result, this project is rated overall as a MODERATE Risk project. During the project design, management measures has been incorporated into ProDoc.
Substantial Risk	☐	
High Risk	☐	

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

	QUESTION 5: Based on the identified risks and risk categorization, what requirements of the SES are triggered? (check all that apply)				
	Question only required for Moderate, Substantial and High Risk projects				
	Is assessment required? (check if "yes")	**X**			***Status? (completed, planned)***
	if yes, indicate overall type and status			Targeted assessment(s)	
				ESIA (Environmental and Social Impact Assessment)	I presume we make at least screening for the HPP Sure. Usually we discuss with the MoE and proceed according to their requirements
				SESA (Strategic Environmental and (Social Assessment)	
	Are management plans required? (check if "yes)				
	If yes, indicate overall type			Targeted management plans (e.g. Gender Action Plan, Emergency Response Plan, Waste Management Plan, others)	

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

		X		ESMP (Environmental and Social Management Plan which may include range of targeted plans)	
		X		ESMF (Environmental and Social Management Framework)	
	Based on identified <u>risks</u>, which Principles/Project-level Standards triggered?		**Comments (not required)**		
	Overarching Principle: Leave No One Behind				
	Human Rights	X			
	Gender Equality and Women's Empowerment	X			
	Accountability	X			
	1. Biodiversity Conservation and Sustainable Natural Resource Management				
	2. Climate Change and Disaster Risks	X			
	3. Community Health, Safety and Security	X			
	4. Cultural Heritage				
	5. Displacement and Resettlement				
	6. Indigenous Peoples				
	7. Labour and Working Conditions	X			
	8. Pollution Prevention and Resource Efficiency	X			

Final Sign Off

Signature	*Date*	*Description*
QA Assessor DocuSigned by: **Jihan Seoud** **Programme Manager** B9C552CD991843C...	22-Mar-2024	UNDP staff member responsible for the project, typically a UNDP Programme Officer. Final signature confirms they have "checked" to ensure that the SESP is adequately conducted.
QA Approver DocuSigned by: **Mohammed Saih** **Resident Representative a.i.** ...BB7AEAA5467...	22-Mar-2024	UNDP senior manager, typically the UNDP Deputy Country Director (DCD), Country Director (CD), Deputy Resident Representative (DRR), or Resident Representative (RR). The QA Approver cannot also be the QA Assessor. Final signature confirms they have "cleared" the SESP prior to submittal to the PAC.
PAC Chair DocuSigned by: **Mohammed Saih** **Resident Representative a.i.** ...BB7AEAA5467...	22-Mar-2024	UNDP chair of the PAC. In some cases PAC Chair may also be the QA Approver. Final signature confirms that the SESP was considered as part of the project appraisal and considered in recommendations of the PAC.

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

SESP Attachment 1. Social and Environmental Risk Screening Checklist

Checklist Potential Social and Environmental Risks		
INSTRUCTIONS: The risk screening checklist will assist in answering Questions 2-6 of the Screening Template. Answers to the checklist questions help to (1) identify potential risks, (2) determine the overall risk categorization of the project, and (3) determine required level of assessment and management measures. Refer to the SES toolkit for further guidance on addressing screening questions.		
Overarching Principle: Leave No One Behind **Human Rights**		**Answer (Yes/No)**
P.1	Have local communities or individuals raised human rights concerns regarding the project (e.g. during the stakeholder engagement process, grievance processes, public statements)?	No
P.2	Is there a risk that duty-bearers (e.g. government agencies) do not have the capacity to meet their obligations in the project?	Yes
P.3	Is there a risk that rights-holders (e.g. project-affected persons) do not have the capacity to claim their rights?	Yes
Would the project potentially involve or lead to:		
P.4	adverse impacts on enjoyment of the human rights (civil, political, economic, social or cultural) of the affected population and particularly of marginalized groups?	No
P.5	inequitable or discriminatory impacts on affected populations, particularly people living in poverty or marginalized or excluded individuals or groups, including persons with disabilities? [16]	Yes
P.6	restrictions in availability, quality of and/or access to resources or basic services, in particular to marginalized individuals or groups, including persons with disabilities?	Yes
P.7	exacerbation of conflicts among and/or the risk of violence to project-affected communities and individuals?	No
Gender Equality and Women's Empowerment		
P.8	Have women's groups/leaders raised gender equality concerns regarding the project, (e.g. during the stakeholder engagement process, grievance processes, public statements)?	No
Would the project potentially involve or lead to:		
P.9	adverse impacts on gender equality and/or the situation of women and girls?	No
P.10	reproducing discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits?	Yes
P.11	limitations on women's ability to use, develop and protect natural resources, taking into account different roles and positions of women and men in accessing environmental goods and services? *For example, activities that could lead to natural resources degradation or depletion in communities who depend on these resources for their livelihoods and well being*	No
P.12	exacerbation of risks of gender-based violence? *For example, through the influx of workers to a community, changes in community and household power dynamics, increased exposure to unsafe public places and/or transport, etc.*	No
Sustainability and Resilience: Screening questions regarding risks associated with sustainability and resilience are encompassed by the Standard-specific questions below		
Accountability		
Would the project potentially involve or lead to:		
P.13	exclusion of any potentially affected stakeholders, in particular marginalized groups and excluded individuals (including persons with disabilities), from fully participating in decisions that may affect them?	Yes

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

P.14	grievances or objections from potentially affected stakeholders?	Yes
P.15	risks of retaliation or reprisals against stakeholders who express concerns or grievances, or who seek to participate in or to obtain information on the project?	No
Project-Level Standards		
Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management		
Would the project potentially involve or lead to:		
1.1	adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or ecosystems and ecosystem services? *For example, through habitat loss, conversion or degradation, fragmentation, hydrological changes*	No
1.2	activities within or adjacent to critical habitats and/or environmentally sensitive areas, including (but not limited to) legally protected areas (e.g. nature reserve, national park), areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local communities?	No
1.3	changes to the use of lands and resources that may have adverse impacts on habitats, ecosystems, and/or livelihoods? (Note: if restrictions and/or limitations of access to lands would apply, refer to Standard 5)	No
1.4	risks to endangered species (e.g. reduction, encroachment on habitat)?	No
1.5	exacerbation of illegal wildlife trade?	No
1.6	introduction of invasive alien species?	No
1.7	adverse impacts on soils?	No
1.8	harvesting of natural forests, plantation development, or reforestation?	No
1.9	significant agricultural production?	No
1.10	animal husbandry or harvesting of fish populations or other aquatic species?	No
1.11	significant extraction, diversion or containment of surface or ground water? *For example, construction of dams, reservoirs, river basin developments, groundwater extraction*	No
1.12	handling or utilization of genetically modified organisms/living modified organisms?[17]	No
1.13	utilization of genetic resources? (e.g. collection and/or harvesting, commercial development)[18]	No
1.14	adverse transboundary or global environmental concerns?	No
Standard 2: Climate Change and Disaster Risks		
Would the project potentially involve or lead to:		
2.1	areas subject to hazards such as earthquakes, floods, landslides, severe winds, storm surges, tsunami or volcanic eruptions?	No
2.2	outputs and outcomes sensitive or vulnerable to potential impacts of climate change or disasters? *For example, through increased precipitation, drought, temperature, salinity, extreme events, earthquakes*	Yes
2.3	increases in vulnerability to climate change impacts or disaster risks now or in the future (also known as maladaptive or negative coping practices)? *For example, changes to land use planning may encourage further development of floodplains, potentially increasing the population's vulnerability to climate change, specifically flooding*	No
2.4	increases of greenhouse gas emissions, black carbon emissions or other drivers of climate change?	No
Standard 3: Community Health, Safety and Security		
Would the project potentially involve or lead to:		
3.1	construction and/or infrastructure development (e.g. roads, buildings, dams)? (Note: the GEF does not finance projects that would involve the construction or rehabilitation of large or complex dams)	No

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

3.2	air pollution, noise, vibration, traffic, injuries, physical hazards, poor surface water quality due to runoff, erosion, sanitation?	No
3.3	harm or losses due to failure of structural elements of the project (e.g. collapse of buildings or infrastructure)?	Yes
3.4	risks of water-borne or other vector-borne diseases (e.g. temporary breeding habitats), communicable and noncommunicable diseases, nutritional disorders, mental health?	No
3.5	transport, storage, and use and/or disposal of hazardous or dangerous materials (e.g. explosives, fuel and other chemicals during construction and operation)?	No
3.6	adverse impacts on ecosystems and ecosystem services relevant to communities' health (e.g. food, surface water purification, natural buffers from flooding)?	No
3.7	influx of project workers to project areas?	No
3.8	engagement of security personnel to protect facilities and property or to support project activities?	No
Standard 4: Cultural Heritage		
Would the project potentially involve or lead to:		
4.1	activities adjacent to or within a Cultural Heritage site?	No
4.2	significant excavations, demolitions, movement of earth, flooding or other environmental changes?	No
4.3	adverse impacts to sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture (e.g. knowledge, innovations, practices)? (Note: projects intended to protect and conserve Cultural Heritage may also have inadvertent adverse impacts)	No
4.4	alterations to landscapes and natural features with cultural significance?	No
4.5	utilization of tangible and/or intangible forms (e.g. practices, traditional knowledge) of Cultural Heritage for commercial or other purposes?	No
Standard 5: Displacement and Resettlement		
Would the project potentially involve or lead to:		
5.1	temporary or permanent and full or partial physical displacement (including people without legally recognizable claims to land)?	No
5.2	economic displacement (e.g. loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?	No
5.3	risk of forced evictions?[19]	No
5.4	impacts on or changes to land tenure arrangements and/or community based property rights/customary rights to land, territories and/or resources?	No
Standard 6: Indigenous Peoples		
Would the project potentially involve or lead to:		
6.1	areas where indigenous peoples are present (including project area of influence)?	No
6.2	activities located on lands and territories claimed by indigenous peoples?	No

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

6.3	impacts (positive or negative) to the human rights, lands, natural resources, territories, and traditional livelihoods of indigenous peoples (regardless of whether indigenous peoples possess the legal titles to such areas, whether the project is located within or outside of the lands and territories inhabited by the affected peoples, or whether the indigenous peoples are recognized as indigenous peoples by the country in question)? *If the answer to screening question 6.3 is "yes", then Standard 6 requirements apply, and the potential significance of risks related to impacts on indigenous peoples must be Moderate or above.* *	No
6.4	the absence of culturally appropriate consultations carried out with the objective of achieving FPIC on matters that may affect the rights and interests, lands, resources, territories and traditional livelihoods of the indigenous peoples concerned?	No
6.5	the utilization and/or commercial development of natural resources on lands and territories claimed by indigenous peoples?	No
6.6	forced eviction or the whole or partial physical or economic displacement of indigenous peoples, including through access restrictions to lands, territories, and resources? *Consider, and where appropriate ensure, consistency with the answers under Standard 5 above*	No
6.7	adverse impacts on the development priorities of indigenous peoples as defined by them?	No
6.8	risks to the physical and cultural survival of indigenous peoples?	No
6.9	impacts on the Cultural Heritage of indigenous peoples, including through the commercialization or use of their traditional knowledge and practices? *Consider, and where appropriate ensure, consistency with the answers under Standard 4 above.*	No
Standard 7: Labour and Working Conditions		
Would the project potentially involve or lead to: (note: applies to project and contractor workers)		
7.1	working conditions that do not meet national labour laws and international commitments?	Yes
7.2	working conditions that may deny freedom of association and collective bargaining?	Yes
7.3	use of child labour?	Yes
7.4	use of forced labour?	Yes
7.5	discriminatory working conditions and/or lack of equal opportunity?	Yes
7.6	occupational health and safety risks due to physical, chemical, biological and psychosocial hazards (including violence and harassment) throughout the project life-cycle?	Yes
Standard 8: Pollution Prevention and Resource Efficiency		
Would the project potentially involve or lead to:		
8.1	the release of pollutants to the environment due to routine or non-routine circumstances with the potential for adverse local, regional, and/or transboundary impacts?	No
8.2	the generation of waste (both hazardous and non-hazardous)?	Yes
8.3	the manufacture, trade, release, and/or use of hazardous materials and/or chemicals?	No
8.4	the use of chemicals or materials subject to international bans or phase-outs? *For example, DDT, PCBs and other chemicals listed in international conventions such as the Montreal Protocol, Minamata Convention, Basel Convention, Rotterdam Convention, Stockholm Convention*	No
8.5	the application of pesticides that may have a negative effect on the environment or human health?	No
8.6	significant consumption of raw materials, energy, and/or water?	No

* Note: revised July 2022 modifying presumption of risk significance from Substantial or higher to Moderate or higher.

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

Annex 3 Risk Analysis

OFFLINE PROJECT RISK REGISTER TEMPLATE

Project Title: Improving energy security and promoting the transition to sustainable energy in Lebanon	Project Number: 01001708	Date: prodoc signature

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
1	There is a risk of damage to installed equipment and material …	As a result of beneficiaries not maintaining or protecting the system	Which will impact in lack of sustainability of equipment/real benefit	**3. OPERATIONAL (3.5. Partners' engagement) - UNDP Risk Appetite: EXPLORATORY TO OPEN**	Likelihood: **3 - Moderately likely** Impact: **3 - Intermediate** Risk level: **MODERATE (equates to a risk appetite of EXPLORATORY)**	From: 06-June 24 To: 30-December-28	UNDP	Risk Treatment 1.1: Memorandums of understanding will be signed with beneficiaries to ensure the protection, operation and maintenance of the equipment installed. Risk Treatment Owner: Project Manager
2	There is a risk that the solar systems will be damaged by	As a result of natural disasters or environmental degradation	Which will impact on their sustainability/operation….	**8.Safety and Security (8.6. Natural hazards) - UNDP**	Likelihood: **1 - Not likely** Impact:	From: 01-Jan 24 To: 30-	… UNDP	Risk Treatment 2.1: Ensure all outdoor equipment can withstand severe weather conditions expected in Lebanon

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

	natural disasters or environmental degredation			**Risk Appetite: CAUTIOUS**	**3 - Intermediate** Risk level: **LOW (equates to a risk appetite of CAUTIOUS)**	December-28		Risk Treatment Owner: Project Manager Risk Treatment 2.2: Ensure that an environmental statement is undertaken for all projects with significant installations…. Risk Treatment Owner: Project Manager
3	There is a risk that at the end of the project the batteries may need to be disposed	As a result of duration/lifetime of the technology	Which will impact in solid waste	**1. SOCIAL AND ENVIRONMENTAL (1.11. Pollution prevention and resource efficiency) - UNDP Risk Appetite: CAUTIOUS**	Likelihood: **3 - Moderately likely** Impact: **3 - Intermediate** Risk level: **MODERATE (equates to a risk appetite ofEXPLORATORY)**	From: 06-Jun-24 To: 30-Dec-28	UNDP	Risk Treatment 3.1: UNDP will be working on battery disposal through the EU-funded TaDWIR project. Activities will be coordinated with this Action. … Risk Treatment Owner: Project Manager

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

4	There is a risk that the beneficiaires will not have the ability to effectively operate and maintain installed systems …	As a result of lack of financing and/or capacity…	Which will impact in the operation of the installed systems….	**3. OPERATIONAL (3.8. Capacities of the partners) - UNDP Risk Appetite: EXPLORATORY TO OPEN**	Likelihood: **4 - Highly likely** Impact: **4 - Extensive** Risk level: **SUBSTANTIAL (equates to a risk appetite of OPEN)**	From: 01-Jun 24 To: 30-Dec-28	… UNDP	Risk Treatment 4.1: A minimum of two-years performance guarantee for all systems installed will be required. The possibility of including O&M contract will also be assessed and capacity building activities will be ensured. Risk Treatment Owner: Project Manager

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

5	There is a risk that delays in import clearance will cause delays in delivering the equipment, thus delaying the implementation of the project activities	As a result of institutional capacity of government entities	Which will impact in delay in project implementation	**3. ORGANIZATIONAL (4.1 Governance) - UNDP Risk Appetite: EXPLORATORY TO OPEN**	Likelihood: **3 - Moderately likely** Impact: **3 - Intermediate** Risk level: **MODERATE (equates to a risk appetite of CAUTIOUS)**	From: 01-Jun-24 To: 30-Dec-28	… *UNDP*	Risk Treatment 5.1: Sufficient time will be given for project implementation to account for expected delays in manufacturing, shipping and clearance of imported system components … Risk Treatment Owner: Project Manager

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

6	There is a risk of delays in manufacturing and shipping of equipment	Lack of planning and coordination - External factors like blocking of shipping routes	Which will impact in…delay in project implementation	**3. OPERATIONAL (3.8. Capacities of the partners) - UNDP Risk Appetite: EXPLORATORY TO OPEN**	Likelihood: **2 - Low likelihood** Impact: **3 - Intermediate** Risk level: **MODERATE (equates to a risk appetite of EXPLORATORY)**	From: 01-Jun-24 To: 30-Dec-28	…	Risk Treatment 6.1: Continuous coordination will be conducted with the contractors and suppliers to overcome any possible delays in manufacturing or shipping.… Risk Treatment Owner: Project Manager

DocuSign Envelope ID: 1DA4EE80-B055-488C-BE16-7423F018EA46

7	There is a risk that there is Instability in the security situation in the country …	As a result of the geopolitical context and regional conflicts	Which will impact the capacity to implement is affected (field work, ...) and delays in implementation	**8. SAFETY AND SECURITY (8.1. Armed conflict) - UNDP Risk Appetite: CAUTIOUS**	Likelihood: **3 - Moderately likely** Impact: **3 - Intermediate** Risk level: **MODERATE (equates to a risk appetite of EXPLORATORY)**	From: 01-Nov-23 To: 30-Dec-28	UNDP	Risk Treatment 7.1: Isolated security incidences may occur. Appropriate measures will be taken accordingly. These could include the postponement of some of the works on site, and the coordination with the Internal Security Forces and UN Department of Security as needed… Risk Treatment Owner: Project Manager